

CHAPTER

# 6 FET Biasing

## 6.1 INTRODUCTION

In Chapter 5 we found that the biasing levels for a silicon transistor configuration can be obtained using the characteristic equations $V_{BE} = 0.7$ V, $I_C = \beta I_B$, and $I_C \cong I_E$. The linkage between input and output variables is provided by $\beta$, which is assumed to be fixed in magnitude for the analysis to be performed. The fact that beta is a constant establishes a *linear* relationship between $I_C$ and $I_B$. Doubling the value of $I_B$ will double the level of $I_C$, and so on.

For the field-effect transistor, the relationship between input and output quantities is *nonlinear* due to the squared term in Shockley's equation. Linear relationships result in straight lines when plotted on a graph of one variable versus the other, while nonlinear functions result in curves as obtained for the transfer characteristics of a JFET. The nonlinear relationship between $I_D$ and $V_{GS}$ can complicate the mathematical approach to the dc analysis of FET configurations. A graphical approach may limit solutions to tenths-place accuracy, but it is a quicker method for most FET amplifiers. Since the graphical approach is in general the most popular, the analysis of this chapter will have a graphical orientation rather than direct mathematical techniques.

Another distinct difference between the analysis of BJT and FET transistors is that the input controlling variable for a BJT transistor is a current level, while for the FET a voltage is the controlling variable. In both cases, however, the controlled variable on the output side is a current level that also defines the important voltage levels of the output circuit.

The general relationships that can be applied to the dc analysis of all FET amplifiers are

$$I_G \cong 0 \text{ A} \tag{6.1}$$

and

$$I_D = I_S \tag{6.2}$$

For JFETS and depletion-type MOSFETs, Shockley's equation is applied to relate the input and output quantities:

$$I_D = I_{DSS}\left(\frac{1 - V_{GS}}{V_P}\right)^2 \tag{6.3}$$

For enhancement-type MOSFETs, the following equation is applicable:

$$I_D = k(V_{GS} - V_T)^2 \tag{6.4}$$

It is particularly important to realize that all of the equations above are for the *device only!* They do not change with each network configuration so long as the device is in the active region. The network simply defines the level of current and voltage associated with the operating point through its own set of equations. In reality, the dc solution of BJT and FET networks is the solution of simultaneous equations established by the device and network. The solution can be determined using a mathematical or graphical approach—a fact to be demonstrated by the first few networks to be analyzed. However, as noted earlier, the graphical approach is the most popular for FET networks and is employed in this book.

The first few sections of this chapter are limited to JFETs and the graphical approach to analysis. The depletion-type MOSFET will then be examined with its increased range of operating points, followed by the enhancement-type MOSFET. Finally, problems of a design nature are investigated to fully test the concepts and procedures introduced in the chapter.

## 6.2 FIXED-BIAS CONFIGURATION

The simplest of biasing arrangements for the *n*-channel JFET appears in Fig. 6.1. Referred to as the fixed-bias configuration, it is one of the few FET configurations that can be solved just as directly using either a mathematical or graphical approach. Both methods are included in this section to demonstrate the difference between the two philosophies and also to establish the fact that the same solution can be obtained using either method.

The configuration of Fig. 6.1 includes the ac levels $V_i$ and $V_o$ and the coupling capacitors ($C_1$ and $C_2$). Recall that the coupling capacitors are "open circuits" for the dc analysis and low impedances (essentially short circuits) for the ac analysis. The resistor $R_G$ is present to ensure that $V_i$ appears at the input to the FET amplifier for the ac analysis (Chapter 9). For the dc analysis,

$$I_G \cong 0 \text{ A}$$

and

$$V_{R_G} = I_G R_G = (0 \text{ A})R_G = 0 \text{ V}$$

The zero-volt drop across $R_G$ permits replacing $R_G$ by a short-circuit equivalent, as appearing in the network of Fig. 6.2 specifically redrawn for the dc analysis.

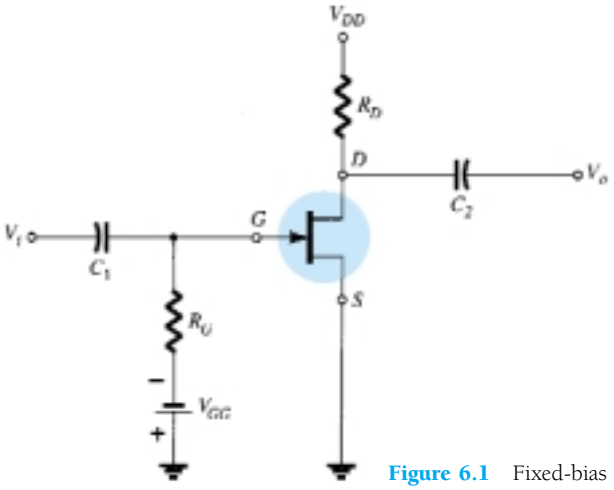



**Figure 6.1** Fixed-bias configuration.

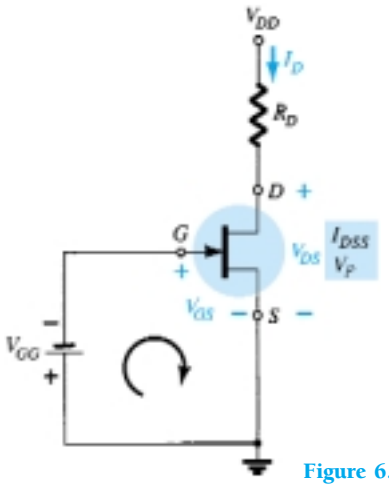



**Figure 6.2** Network for dc analysis.

The fact that the negative terminal of the battery is connected directly to the defined positive potential of $V_{GS}$ clearly reveals that the polarity of $V_{GS}$ is directly opposite to that of $V_{GG}$. Applying Kirchhoff's voltage law in the clockwise direction of the indicated loop of Fig. 6.2 will result in

$$-V_{GG} - V_{GS} = 0$$

and

$$\boxed{V_{GS} = -V_{GG}} \tag{6.5}$$

Since $V_{GG}$ is a fixed dc supply, the voltage $V_{GS}$ is fixed in magnitude, resulting in the notation "fixed-bias configuration."

The resulting level of drain current $I_D$ is now controlled by Shockley's equation:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

Since $V_{GS}$ is a fixed quantity for this configuration, its magnitude and sign can simply be substituted into Shockley's equation and the resulting level of $I_D$ calculated. This is one of the few instances in which a mathematical solution to a FET configuration is quite direct.

A graphical analysis would require a plot of Shockley's equation as shown in Fig. 6.3. Recall that choosing $V_{GS} = V_P/2$ will result in a drain current of $I_{DSS}/4$ when plotting the equation. For the analysis of this chapter, the three points defined by $I_{DSS}$, $V_P$, and the intersection just described will be sufficient for plotting the curve.

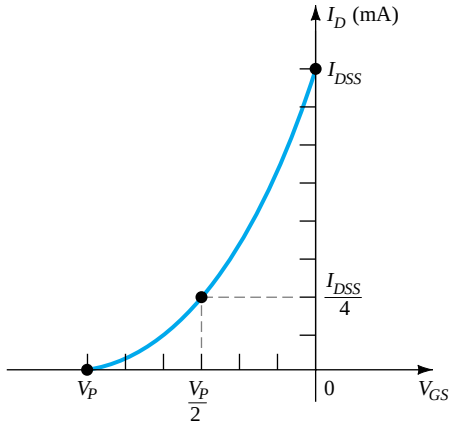



**Figure 6.3** Plotting Shockley's equation.

In Fig. 6.4, the fixed level of $V_{GS}$ has been superimposed as a vertical line at $V_{GS} = -V_{GG}$. At any point on the vertical line, the level of $V_{GS}$ is $-V_{GG}$—the level of $I_D$ must simply be determined on this vertical line. The point where the two curves

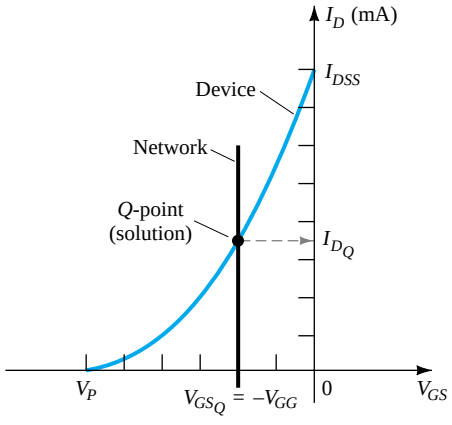



**Figure 6.4** Finding the solution for the fixed-bias configuration.

intersect is the common solution to the configuration—commonly referred to as the *quiescent* or *operating point.* The subscript $Q$ will be applied to drain current and gate-to-source voltage to identify their levels at the $Q$-point. Note in Fig. 6.4 that the quiescent level of $I_D$ is determined by drawing a horizontal line from the $Q$-point to the vertical $I_D$ axis as shown in Fig. 6.4. It is important to realize that once the network of Fig. 6.1 is constructed and operating, the dc levels of $I_D$ and $V_{GS}$ that will be measured by the meters of Fig. 6.5 are the quiescent values defined by Fig. 6.4.

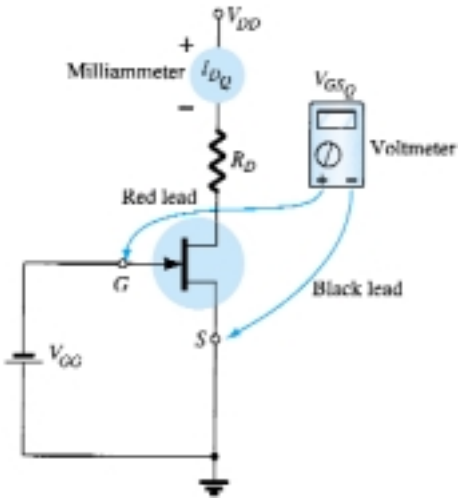



**Figure 6.5** Measuring the quiescent values of $I_D$ and $V_{GS}$.

The drain-to-source voltage of the output section can be determined by applying Kirchhoff's voltage law as follows:

$$+V_{DS} + I_D R_D - V_{DD} = 0$$

and

$$\boxed{V_{DS} = V_{DD} - I_D R_D} \tag{6.6}$$

Recall that single-subscript voltages refer to the voltage at a point with respect to ground. For the configuration of Fig. 6.2,

$$\boxed{V_S = 0\text{ V}} \tag{6.7}$$

Using double-subscript notation:

$$V_{DS} = V_D - V_S$$

or

$$V_D = V_{DS} + V_S = V_{DS} + 0\text{ V}$$

and

$$\boxed{V_D = V_{DS}} \tag{6.8}$$

In addition,

$$V_{GS} = V_G - V_S$$

or

$$V_G = V_{GS} + V_S = V_{GS} + 0\text{ V}$$

and

$$\boxed{V_G = V_{GS}} \tag{6.9}$$

The fact that $V_D = V_{DS}$ and $V_G = V_{GS}$ is fairly obvious from the fact that $V_S = 0$ V, but the derivations above were included to emphasize the relationship that exists between double-subscript and single-subscript notation. Since the configuration requires two dc supplies, its use is limited and will not be included in the forthcoming list of the most common FET configurations.

## EXAMPLE 6.1

Determine the following for the network of Fig. 6.6.

(a) $V_{GSQ}$.
(b) $I_{DQ}$.
(c) $V_{DS}$.
(d) $V_D$.
(e) $V_G$.
(f) $V_S$.

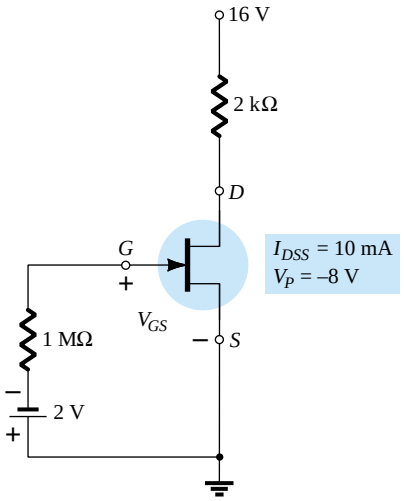



**Figure 6.6** Example 6.1.

### Solution

**Mathematical Approach:**

(a) $V_{GSQ} = -V_{GG} = \mathbf{-2\ V}$

(b) $$I_{DQ} = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 10\text{ mA}\left(1 - \frac{-2\text{ V}}{-8\text{ V}}\right)^2$$
$$= 10\text{ mA}(1 - 0.25)^2 = 10\text{ mA}(0.75)^2 = 10\text{ mA}(0.5625)$$
$$= \mathbf{5.625\ mA}$$

(c) $$V_{DS} = V_{DD} - I_D R_D = 16\text{ V} - (5.625\text{ mA})(2\text{ k}\Omega)$$
$$= 16\text{ V} - 11.25\text{ V} = \mathbf{4.75\ V}$$

(d) $V_D = V_{DS} = \mathbf{4.75\ V}$

(e) $V_G = V_{GS} = \mathbf{-2\ V}$

(f) $V_S = \mathbf{0\ V}$

**Graphical Approach:**

The resulting Shockley curve and the vertical line at $V_{GS} = -2$ V are provided in Fig. 6.7. It is certainly difficult to read beyond the second place without significantly in-

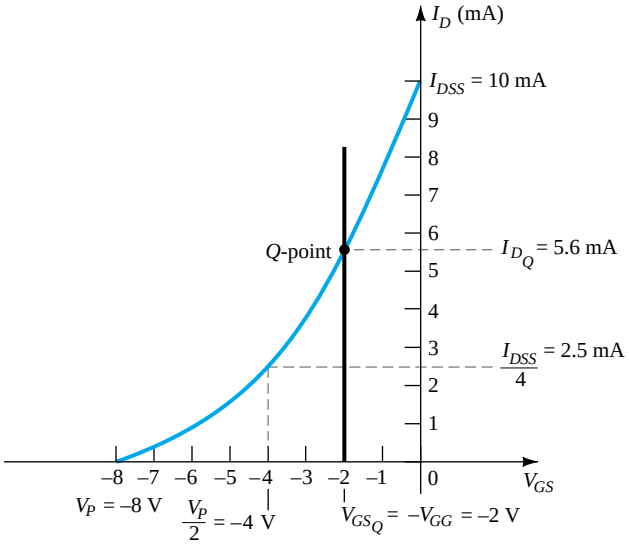



**Figure 6.7** Graphical solution for the network of Fig. 6.6.

creasing the size of the figure, but a solution of 5.6 mA from the graph of Fig. 6.7 is quite acceptable. Therefore, for part (a),

$$V_{GS_Q} = -V_{GG} = \mathbf{-2\ V}$$

(b) $I_{D_Q} = \mathbf{5.6\ mA}$

(c) $V_{DS} = V_{DD} - I_D R_D = 16\ \text{V} - (5.6\ \text{mA})(2\ \text{k}\Omega)$
$= 16\ \text{V} - 11.2\ \text{V} = \mathbf{4.8\ V}$

(d) $V_D = V_{DS} = \mathbf{4.8\ V}$

(e) $V_G = V_{GS} = \mathbf{-2\ V}$

(f) $V_S = \mathbf{0\ V}$

The results clearly confirm the fact that the mathematical and graphical approaches generate solutions that are quite close.

## 6.3 SELF-BIAS CONFIGURATION

The self-bias configuration eliminates the need for two dc supplies. The controlling gate-to-source voltage is now determined by the voltage across a resistor $R_S$ introduced in the source leg of the configuration as shown in Fig. 6.8.

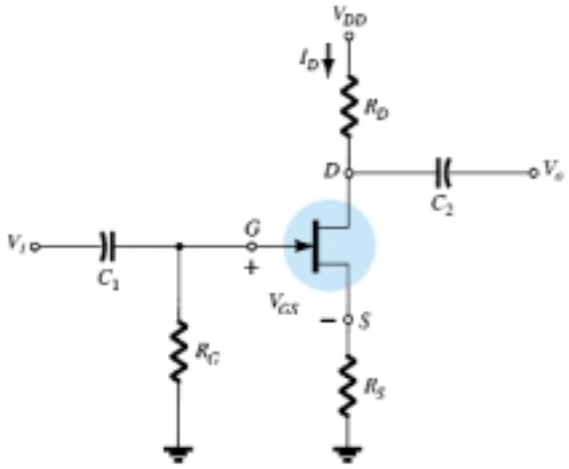



**Figure 6.8** JFET self-bias configuration.

For the dc analysis, the capacitors can again be replaced by "open circuits" and the resistor $R_G$ replaced by a short-circuit equivalent since $I_G = 0$ A. The result is the network of Fig. 6.9 for the important dc analysis.

The current through $R_S$ is the source current $I_S$, but $I_S = I_D$ and

$$V_{R_s} = I_D R_S$$

For the indicated closed loop of Fig. 6.9, we find that

$$-V_{GS} - V_{R_s} = 0$$

and

$$V_{GS} = -V_{R_s}$$

or

$$\boxed{V_{GS} = -I_D R_S} \tag{6.10}$$

Note in this case that $V_{GS}$ is a function of the output current $I_D$ and not fixed in magnitude as occurred for the fixed-bias configuration.

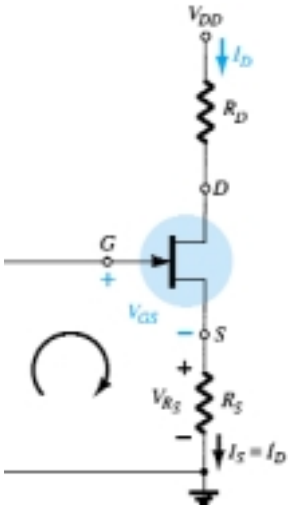



**Figure 6.9** DC analysis of the self-bias configuration.

Equation (6.10) is defined by the network configuration, and Shockley's equation relates the input and output quantities of the device. Both equations relate the same two variables, permitting either a mathematical or graphical solution.

A mathematical solution could be obtained simply by substituting Eq. (6.10) into Shockley's equation as shown below:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= I_{DSS}\left(1 - \frac{-I_D R_S}{V_P}\right)^2$$

or

$$I_D = I_{DSS}\left(1 + \frac{I_D R_S}{V_P}\right)^2$$

By performing the squaring process indicated and rearranging terms, an equation of the following form can be obtained:

$$I_D^2 + K_1 I_D + K_2 = 0$$

The quadratic equation can then be solved for the appropriate solution for $I_D$.

The sequence above defines the mathematical approach. The graphical approach requires that we first establish the device transfer characteristics as shown in Fig. 6.10. Since Eq. (6.10) defines a straight line on the same graph, let us now identify two points on the graph that are on the line and simply draw a straight line between the two points. The most obvious condition to apply is $I_D = 0$ A since it results in $V_{GS} = -I_D R_S = (0 \text{ A}) R_S = 0$ V. For Eq. (6.10), therefore, one point on the straight line is defined by $I_D = 0$ A and $V_{GS} = 0$ V, as appearing on Fig. 6.10.

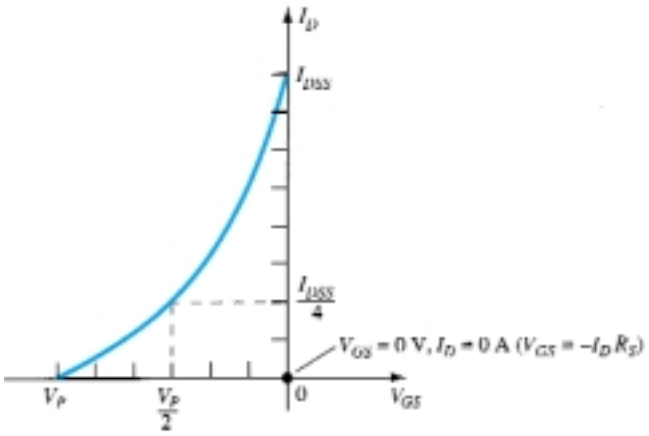



**Figure 6.10** Defining a point on the self-bias line.

The second point for Eq. (6.10) requires that a level of $V_{GS}$ or $I_D$ be chosen and the corresponding level of the other quantity be determined using Eq. (6.10). The resulting levels of $I_D$ and $V_{GS}$ will then define another point on the straight line and permit an actual drawing of the straight line. Suppose, for example, that we choose a level of $I_D$ equal to one-half the saturation level. That is,

$$I_D = \frac{I_{DSS}}{2}$$

then

$$V_{GS} = -I_D R_S = -\frac{I_{DSS} R_S}{2}$$

The result is a second point for the straight-line plot as shown in Fig. 6.11. The straight line as defined by Eq. (6.10) is then drawn and the quiescent point obtained at the in-

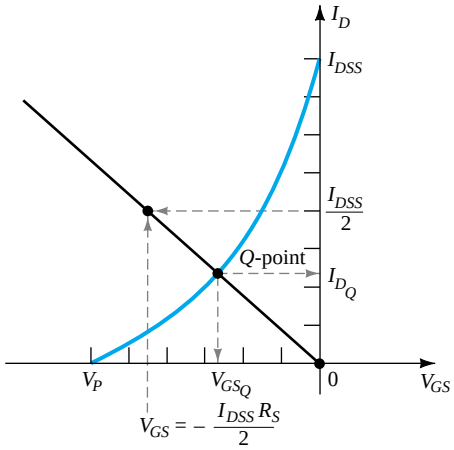



**Figure 6.11** Sketching the self-bias line.

tersection of the straight-line plot and the device characteristic curve. The quiescent values of $I_D$ and $V_{GS}$ can then be determined and used to find the other quantities of interest.

The level of $V_{DS}$ can be determined by applying Kirchhoff's voltage law to the output circuit, with the result that

$$V_{R_S} + V_{DS} + V_{R_D} - V_{DD} = 0$$

and
$$V_{DS} = V_{DD} - V_{R_S} - V_{R_D} = V_{DD} - I_S R_S - I_D R_D$$

but
$$I_D = I_S$$

and
$$\boxed{V_{DS} = V_{DD} - I_D(R_S + R_D)} \tag{6.11}$$

In addition:

$$\boxed{V_S = I_D R_S} \tag{6.12}$$

$$\boxed{V_G = 0\ \text{V}} \tag{6.13}$$

and
$$\boxed{V_D = V_{DS} + V_S = V_{DD} - V_{R_D}} \tag{6.14}$$

## *EXAMPLE 6.2*

Determine the following for the network of Fig. 6.12.

(a) $V_{GS_Q}$.
(b) $I_{D_Q}$.
(c) $V_{DS}$.
(d) $V_S$.
(e) $V_G$.
(f ) $V_D$.

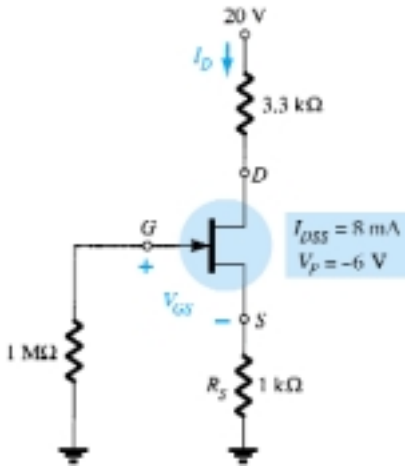



**Figure 6.12** Example 6.2.

## Solution

(a) The gate-to-source voltage is determined by

$$V_{GS} = -I_D R_S$$

Choosing $I_D = 4$ mA, we obtain

$$V_{GS} = -(4 \text{ mA})(1 \text{ k}\Omega) = -4 \text{ V}$$

The result is the plot of Fig. 6.13 as defined by the network.

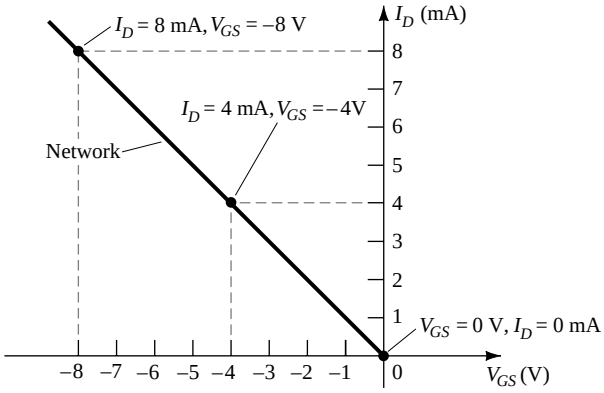



**Figure 6.13** Sketching the self-bias line for the network of Fig. 6.12.

If we happen to choose $I_D = 8$ mA, the resulting value of $V_{GS}$ would be $-8$ V, as shown on the same graph. In either case, the same straight line will result, clearly demonstrating that any appropriate value of $I_D$ can be chosen as long as the corresponding value of $V_{GS}$ as determined by Eq. (6.10) is employed. In addition, keep in mind that the value of $V_{GS}$ could be chosen and the value of $I_D$ calculated with the same resulting plot.

For Shockley's equation, if we choose $V_{GS} = V_P/2 = -3$ V, we find that $I_D = I_{DSS}/4 = 8$ mA/4 $= 2$ mA, and the plot of Fig. 6.14 will result, representing the characteristics of the device. The solution is obtained by superimposing the network characteristics defined by Fig. 6.13 on the device characteristics of Fig. 6.14 and finding the point of intersection of the two as indicated on Fig. 6.15. The resulting operating point results in a quiescent value of gate-to-source voltage of

$$V_{GS_Q} = \mathbf{-2.6 \text{ V}}$$

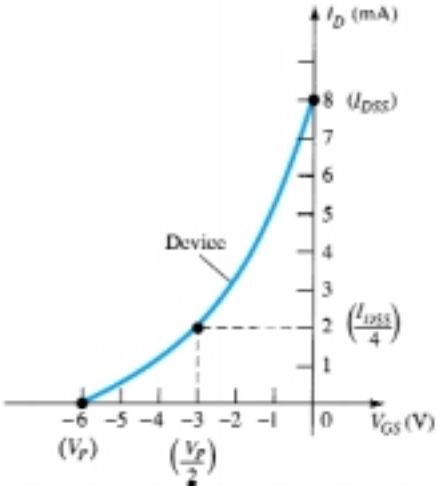



**Figure 6.14** Sketching the device characteristics for the JFET of Fig. 6.12.

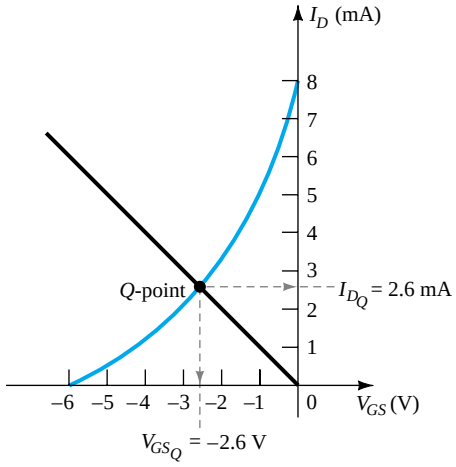



**Figure 6.15** Determining the *Q*-point for the network of Fig. 6.12.

(b) At the quiescent point:

$$I_{DQ} = \mathbf{2.6\ mA}$$

(c) Eq. (6.11): $V_{DS} = V_{DD} - I_D(R_S + R_D)$
$= 20\ \text{V} - (2.6\ \text{mA})(1\ \text{k}\Omega + 3.3\ \text{k}\Omega)$
$= 20\ \text{V} - 11.18\ \text{V}$
$= \mathbf{8.82\ V}$

(d) Eq. (6.12): $V_S = I_D R_S$
$= (2.6\ \text{mA})(1\ \text{k}\Omega)$
$= \mathbf{2.6\ V}$

(e) Eq. (6.13): $V_G = \mathbf{0\ V}$

(f) Eq. (6.14): $V_D = V_{DS} + V_S = 8.82\ \text{V} + 2.6\ \text{V} = \mathbf{11.42\ V}$

or $V_D = V_{DD} - I_D R_D = 20\ \text{V} - (2.6\ \text{mA})(3.3\ \text{k}\Omega) = \mathbf{11.42\ V}$

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*EXAMPLE 6.3*

Find the quiescent point for the network of Fig. 6.12 if:
(a) $R_S = 100\ \Omega$.
(b) $R_S = 10\ \text{k}\Omega$.

**Solution**

Note Fig. 6.16.

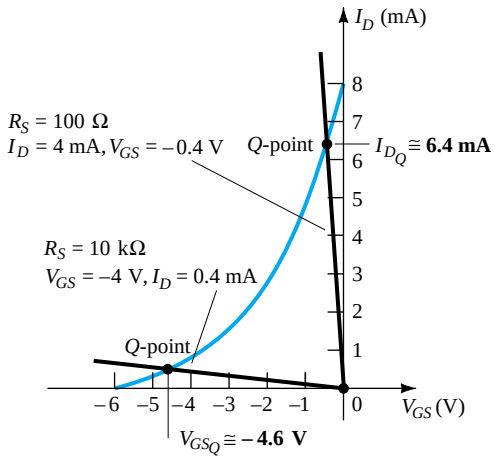



**Figure 6.16** Example 6.3.

(a) With the $I_D$ scale,

$$I_{DQ} \cong \mathbf{6.4\ mA}$$

From Eq. (6.10),

$$V_{GS_Q} \cong \mathbf{-0.64\ V}$$

(b) With the $V_{GS}$ scale,

$$V_{GS_Q} \cong \mathbf{-4.6\ V}$$

From Eq. (6.10),

$$I_{DQ} \cong \mathbf{0.46\ mA}$$

In particular, note how lower levels of $R_S$ bring the load line of the network closer to the $I_D$ axis while increasing levels of $R_S$ bring the load line closer to the $V_{GS}$ axis.

## EXAMPLE 6.4

Determine the following for the common-gate configuration of Fig. 6.17.

(a) $V_{GSQ}$.
(b) $I_{DQ}$.
(c) $V_D$.
(d) $V_G$.
(e) $V_S$.
(f) $V_{DS}$.

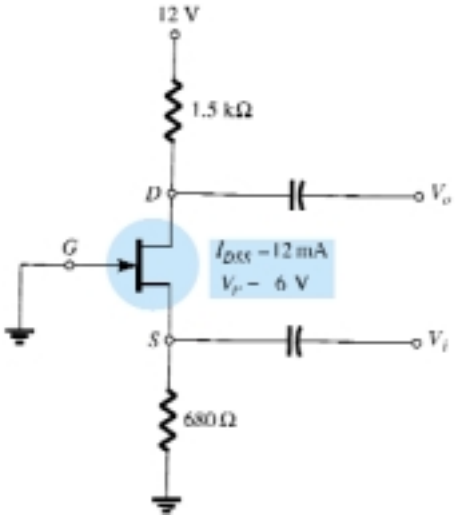



**Figure 6.17** Example 6.4.

### Solution

The grounded gate terminal and the location of the input establish strong similarities with the common-base BJT amplifier. Although different in appearance from the basic structure of Fig. 6.8, the resulting dc network of Fig. 6.18 has the same basic structure as Fig. 6.9. The dc analysis can therefore proceed in the same manner as recent examples.

(a) The transfer characteristics and load line appear in Fig. 6.19. In this case, the second point for the sketch of the load line was determined by choosing (arbitrarily) $I_D = 6$ mA and solving for $V_{GS}$. That is,

$$V_{GS} = -I_D R_S = -(6\text{ mA})(680\ \Omega) = -4.08\text{ V}$$

as shown in Fig. 6.19. The device transfer curve was sketched using

$$I_D = \frac{I_{DSS}}{4} = \frac{12\text{ mA}}{4} = 3\text{ mA}$$

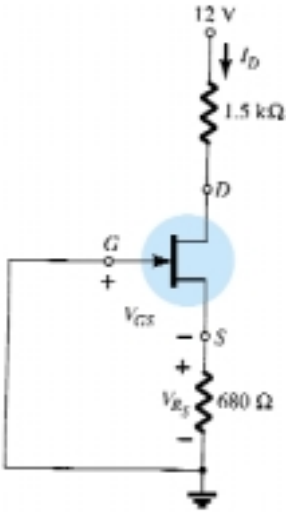



**Figure 6.18** Sketching the dc equivalent of the network of Fig. 6.17.

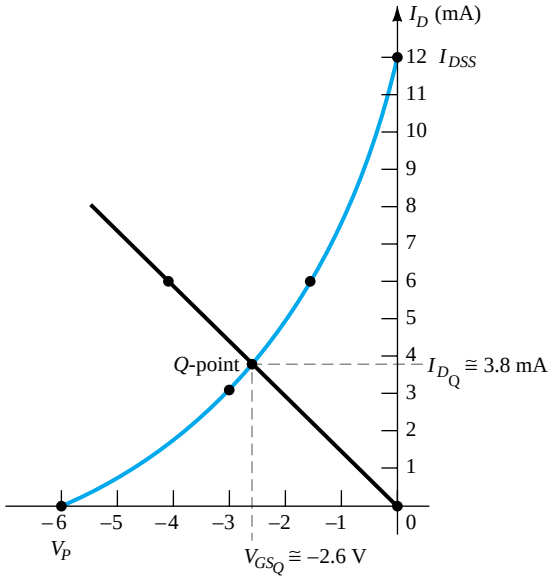



**Figure 6.19** Determining the *Q*-point for the network of Fig. 6.17.

and the associated value of $V_{GS}$:

$$V_{GS} = \frac{V_P}{2} = -\frac{6\text{ V}}{2} = -3\text{ V}$$

as shown on Fig. 6.19. Using the resulting quiescent point of Fig. 6.19 results in

$$V_{GSQ} \cong \mathbf{-2.6\text{ V}}$$

(b) From Fig. 6.19,

$$I_{DQ} \cong \mathbf{3.8\text{ mA}}$$

(c) $V_D = V_{DD} - I_D R_D$
$= 12\text{ V} - (3.8\text{ mA})(1.5\text{ k}\Omega) = 12\text{ V} - 5.7\text{ V}$
$= \mathbf{6.3\text{ V}}$

(d) $V_G = \mathbf{0\text{ V}}$

(e) $V_S = I_D R_S = (3.8\text{ mA})(680\ \Omega)$
$= \mathbf{2.58\text{ V}}$

(f) $V_{DS} = V_D - V_S$
$= 6.3\text{ V} - 2.58\text{ V}$
$= \mathbf{3.72\text{ V}}$

## 6.4 VOLTAGE-DIVIDER BIASING

The voltage-divider bias arrangement applied to BJT transistor amplifiers is also applied to FET amplifiers as demonstrated by Fig. 6.20. The basic construction is exactly the same, but the dc analysis of each is quite different. $I_G = 0$ A for FET amplifiers, but the magnitude of $I_B$ for common-emitter BJT amplifiers can affect the dc levels of current and voltage in both the input and output circuits. Recall that $I_B$ provided the link between input and output circuits for the BJT voltage-divider configuration while $V_{GS}$ will do the same for the FET configuration.

The network of Fig. 6.20 is redrawn as shown in Fig. 6.21 for the dc analysis. Note that all the capacitors, including the bypass capacitor $C_S$, have been replaced by an "open-circuit" equivalent. In addition, the source $V_{DD}$ was separated into two equiv-

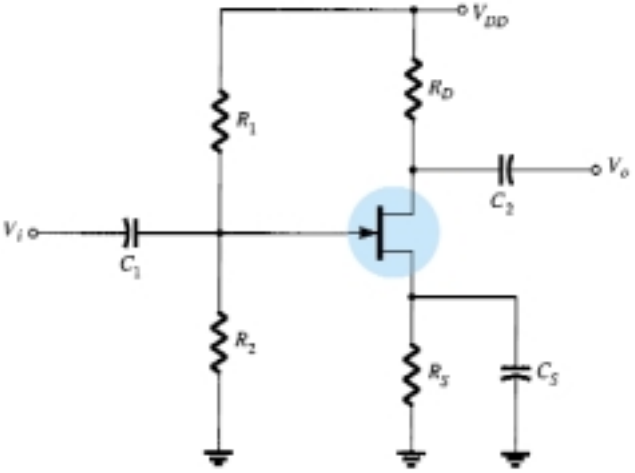



**Figure 6.20** Voltage-divider bias arrangement.

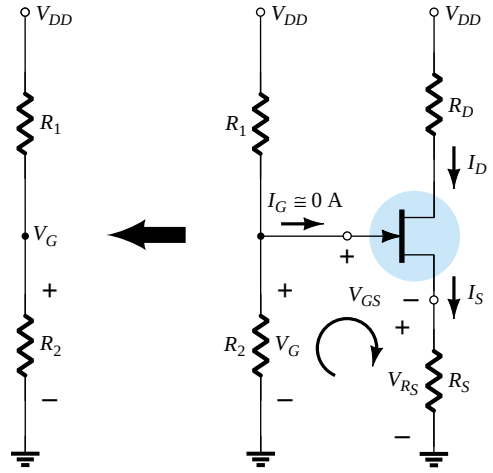



**Figure 6.21** Redrawn network of Fig. 6.20 for dc analysis.

alent sources to permit a further separation of the input and output regions of the network. Since $I_G = 0$ A, Kirchhoff's current law requires that $I_{R_1} = I_{R_2}$ and the series equivalent circuit appearing to the left of the figure can be used to find the level of $V_G$. The voltage $V_G$, equal to the voltage across $R_2$, can be found using the voltage-divider rule as follows:

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} \tag{6.15}$$

Applying Kirchhoff's voltage law in the clockwise direction to the indicated loop of Fig. 6.21 will result in

$$V_G - V_{GS} - V_{R_S} = 0$$

and

$$V_{GS} = V_G - V_{R_S}$$

Substituting $V_{R_S} = I_S R_S = I_D R_S$, we have

$$V_{GS} = V_G - I_D R_S \tag{6.16}$$

The result is an equation that continues to include the same two variables appearing in Shockley's equation: $V_{GS}$ and $I_D$. The quantities $V_G$ and $R_S$ are fixed by the network construction. Equation (6.16) is still the equation for a straight line, but the origin is no longer a point in the plotting of the line. The procedure for plotting Eq. (6.16) is not a difficult one and will proceed as follows. Since any straight line requires two points to be defined, let us first use the fact that *anywhere on the horizontal axis* of Fig. 6.22 the current $I_D = 0$ mA. If we therefore *select* $I_D$ to be 0 mA, we are in essence stating that we are somewhere on the horizontal axis. The exact location can be determined simply by substituting $I_D = 0$ mA into Eq. (6.16) and finding the resulting value of $V_{GS}$ as follows:

$$V_{GS} = V_G - I_D R_S$$
$$= V_G - (0 \text{ mA}) R_S$$

and

$$V_{GS} = V_G\big|_{I_D = 0 \text{ mA}} \tag{6.17}$$

The result specifies that whenever we plot Eq. (6.16), if we choose $I_D = 0$ mA, the value of $V_{GS}$ for the plot will be $V_G$ volts. The point just determined appears in Fig. 6.22.

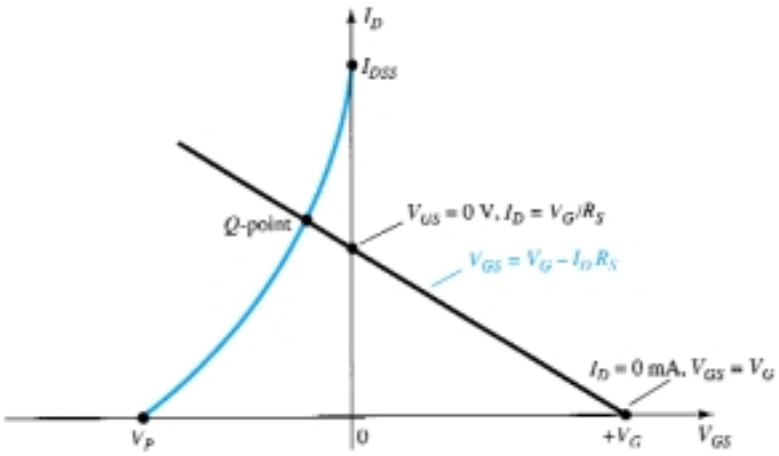



**Figure 6.22** Sketching the network equation for the voltage-divider configuration.

For the other point, let us now employ the fact that at any point on the vertical axis $V_{GS} = 0$ V and solve for the resulting value of $I_D$:

$$V_{GS} = V_G - I_D R_S$$

$$0 \text{ V} = V_G - I_D R_S$$

and

$$\boxed{I_D = \frac{V_G}{R_S}\bigg|_{V_{GS} = 0 \text{ V}}} \tag{6.18}$$

The result specifies that whenever we plot Eq. (6.16), if $V_{GS} = 0$ V, the level of $I_D$ is determined by Eq. (6.18). This intersection also appears on Fig. 6.22.

The two points defined above permit the drawing of a straight line to represent Eq. (6.16). The intersection of the straight line with the transfer curve in the region to the left of the vertical axis will define the operating point and the corresponding levels of $I_D$ and $V_{GS}$.

Since the intersection on the vertical axis is determined by $I_D = V_G/R_S$ and $V_G$ is fixed by the input network, increasing values of $R_S$ will reduce the level of the $I_D$ intersection as shown in Fig. 6.23. It is fairly obvious from Fig. 6.23 that:

*Increasing values of $R_S$ result in lower quiescent values of $I_D$ and more negative values of $V_{GS}$.*

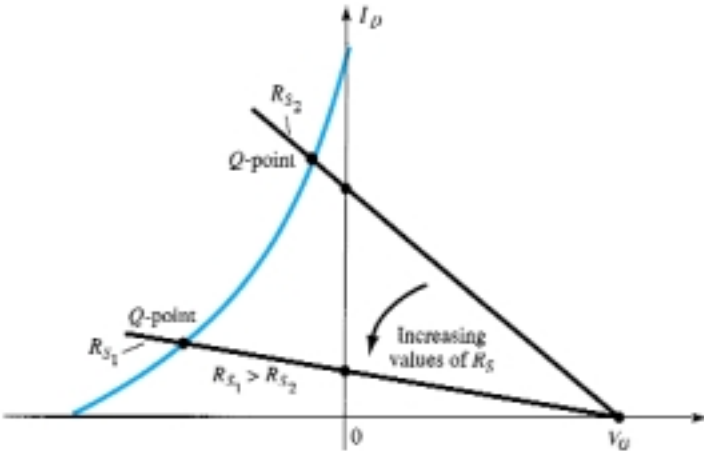



**Figure 6.23** Effect of $R_S$ on the resulting $Q$-point.

Once the quiescent values of $I_{DQ}$ and $V_{GSQ}$ are determined, the remaining network analysis can be performed in the usual manner. That is,

$$\boxed{V_{DS} = V_{DD} - I_D(R_D + R_S)} \tag{6.19}$$

$$\boxed{V_D = V_{DD} - I_D R_D} \tag{6.20}$$

$$\boxed{V_S = I_D R_S} \tag{6.21}$$

$$\boxed{I_{R_1} = I_{R_2} = \frac{V_{DD}}{R_1 + R_2}} \tag{6.22}$$

## EXAMPLE 6.5

Determine the following for the network of Fig. 6.24.

(a) $I_{DQ}$ and $V_{GSQ}$.
(b) $V_D$.
(c) $V_S$.
(d) $V_{DS}$.
(e) $V_{DG}$.

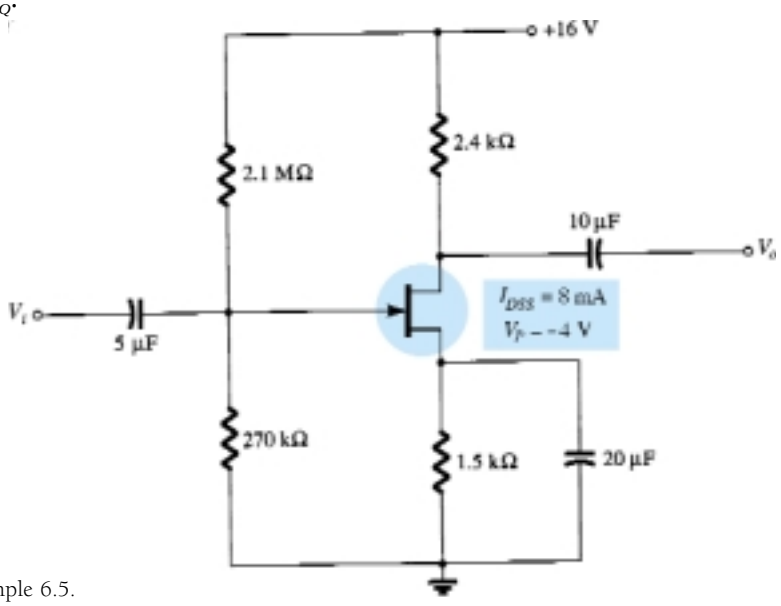



**Figure 6.24** Example 6.5.

### Solution

(a) For the transfer characteristics, if $I_D = I_{DSS}/4 = 8\text{ mA}/4 = 2\text{ mA}$, then $V_{GS} = V_P/2 = -4\text{ V}/2 = -2\text{ V}$. The resulting curve representing Shockley's equation appears in Fig. 6.25. The network equation is defined by

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

$$= \frac{(270\text{ k}\Omega)(16\text{ V})}{2.1\text{ M}\Omega + 0.27\text{ M}\Omega}$$

$$= 1.82\text{ V}$$

and

$$V_{GS} = V_G - I_D R_S$$

$$= 1.82\text{ V} - I_D(1.5\text{ k}\Omega)$$

When $I_D = 0$ mA:

$$V_{GS} = +1.82\text{ V}$$

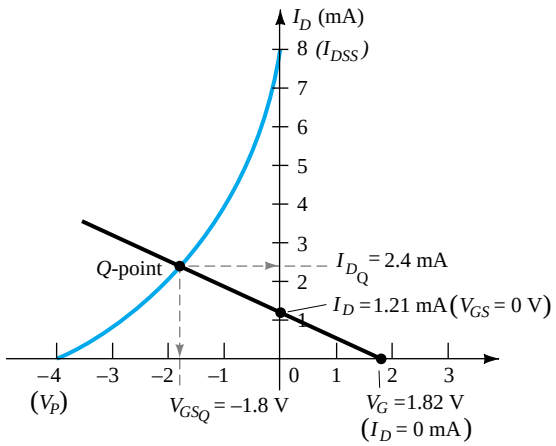



**Figure 6.25** Determining the *Q*-point for the network of Fig. 6.24.

When $V_{GS} = 0$ V:

$$I_D = \frac{1.82 \text{ V}}{1.5 \text{ k}\Omega} = 1.21 \text{ mA}$$

The resulting bias line appears on Fig. 6.25 with quiescent values of

$$I_{D_Q} = \mathbf{2.4 \text{ mA}}$$

and

$$V_{GS_Q} = \mathbf{-1.8 \text{ V}}$$

(b) $V_D = V_{DD} - I_D R_D$
$= 16 \text{ V} - (2.4 \text{ mA})(2.4 \text{ k}\Omega)$
$= \mathbf{10.24 \text{ V}}$

(c) $V_S = I_D R_S = (2.4 \text{ mA})(1.5 \text{ k}\Omega)$
$= \mathbf{3.6 \text{ V}}$

(d) $V_{DS} = V_{DD} - I_D(R_D + R_S)$
$= 16 \text{ V} - (2.4 \text{ mA})(2.4 \text{ k}\Omega + 1.5 \text{ k}\Omega)$
$= \mathbf{6.64 \text{ V}}$
or $V_{DS} = V_D - V_S = 10.24 \text{ V} - 3.6 \text{ V}$
$= \mathbf{6.64 \text{ V}}$

(e) Although seldom requested, the voltage $V_{DG}$ can easily be determined using

$$V_{DG} = V_D - V_G$$
$$= 10.24 \text{ V} - 1.82 \text{ V}$$
$$= \mathbf{8.42 \text{ V}}$$

Although the basic construction of the network in the next example is quite different from the voltage-divider bias arrangement, the resulting equations require a solution very similar to that just described. Note that the network employs a supply at the drain and source.

## *EXAMPLE 6.6*

Determine the following for the network of Fig. 6.26.
(a) $I_{D_Q}$ and $V_{GS_Q}$.
(b) $V_{DS}$.
(c) $V_D$.
(d) $V_S$.

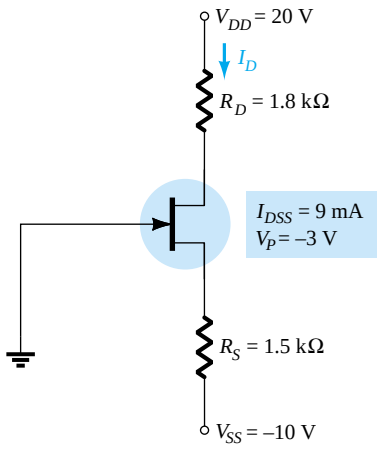



**Figure 6.26** Example 6.6.

## Solution

(a) An equation for $V_{GS}$ in terms of $I_D$ is obtained by applying Kirchhoff's voltage law to the input section of the network as redrawn in Fig. 6.27.

$$-V_{GS} - I_S R_S + V_{SS} = 0$$

or
$$V_{GS} = V_{SS} - I_S R_S$$

but
$$I_S = I_D$$

and
$$\boxed{V_{GS} = V_{SS} - I_D R_S} \tag{6.23}$$

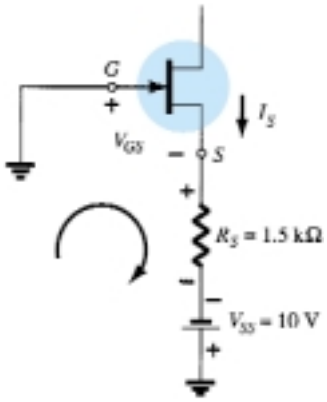



**Figure 6.27** Determining the network equation for the configuration of Fig. 6.26.

The result is an equation very similar in format to Eq. (6.16) that can be superimposed on the transfer characteristics using the procedure described for Eq. (6.16). That is, for this example,

$$V_{GS} = 10\ \text{V} - I_D(1.5\ \text{k}\Omega)$$

For $I_D = 0$ mA,

$$V_{GS} = V_{SS} = 10\ \text{V}$$

For $V_{GS} = 0$ V,

$$0 = 10\ \text{V} - I_D(1.5\ \text{k}\Omega)$$

and
$$I_D = \frac{10\ \text{V}}{1.5\ \text{k}\Omega} = 6.67\ \text{mA}$$

The resulting plot points are identified on Fig. 6.28.

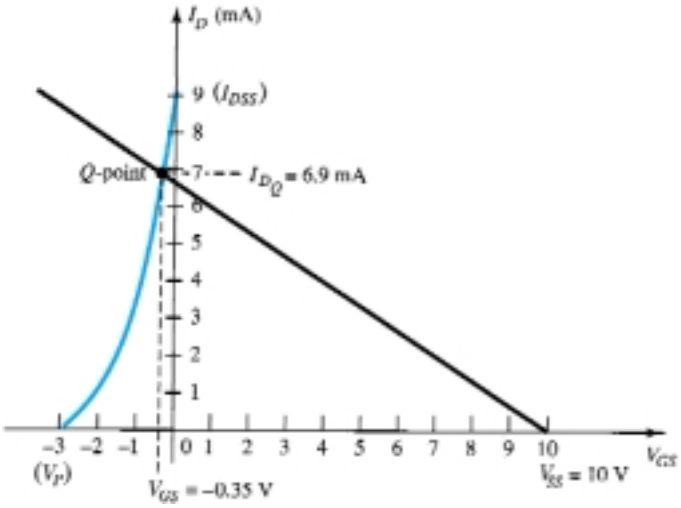



**Figure 6.28** Determining the *Q*-point for the network of Fig. 6.26.

The transfer characteristics are sketched using the plot point established by $V_{GS} = V_P/2 = -3\ \text{V}/2 = -1.5\ \text{V}$ and $I_D = I_{DSS}/4 = 9\ \text{mA}/4 = 2.25\ \text{mA}$, as also appearing on Fig. 6.28. The resulting operating point establishes the following quiescent levels:

$$I_{D_Q} = \mathbf{6.9\ mA}$$
$$V_{GS_Q} = \mathbf{-0.35\ V}$$

(b) Applying Kirchhoff's voltage law to the output side of Fig. 6.26 will result in

$$-V_{SS} + I_S R_S + V_{DS} + I_D R_D - V_{DD} = 0$$

Substituting $I_S = I_D$ and rearranging gives

$$V_{DS} = V_{DD} + V_{SS} - I_D(R_D + R_S) \tag{6.24}$$

which for this example results in

$$V_{DS} = 20\text{ V} + 10\text{ V} - (6.9\text{ mA})(1.8\text{ k}\Omega + 1.5\text{ k}\Omega)$$
$$= 30\text{ V} - 22.77\text{ V}$$
$$= \mathbf{7.23\text{ V}}$$

(c) $V_D = V_{DD} - I_D R_D$
$= 20\text{ V} - (6.9\text{ mA})(1.8\text{ k}\Omega) = 20\text{ V} - 12.42\text{ V}$
$= \mathbf{7.58\text{ V}}$

(d) $V_{DS} = V_D - V_S$
or $V_S = V_D - V_{DS}$
$= 7.58\text{ V} - 7.23\text{ V}$
$= \mathbf{0.35\text{ V}}$

## 6.5 DEPLETION-TYPE MOSFETs

The similarities in appearance between the transfer curves of JFETs and depletion-type MOSFETs permit a similar analysis of each in the dc domain. The primary difference between the two is the fact that depletion-type MOSFETs permit operating points with positive values of $V_{GS}$ and levels of $I_D$ that exceed $I_{DSS}$. In fact, for all the configurations discussed thus far, the analysis is the same if the JFET is replaced by a depletion-type MOSFET.

The only undefined part of the analysis is how to plot Shockley's equation for positive values of $V_{GS}$. How far into the region of positive values of $V_{GS}$ and values of $I_D$ greater than $I_{DSS}$ does the transfer curve have to extend? For most situations, this required range will be fairly well defined by the MOSFET parameters and the resulting bias line of the network. A few examples will reveal the impact of the change in device on the resulting analysis.

*EXAMPLE 6.7*

For the *n*-channel depletion-type MOSFET of Fig. 6.29, determine:
(a) $I_{DQ}$ and $V_{GSQ}$.
(b) $V_{DS}$.

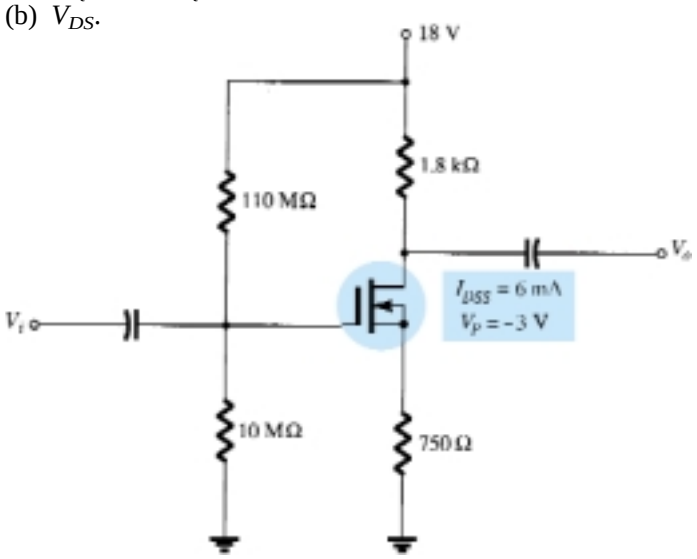



**Figure 6.29** Example 6.7.

## Solution

(a) For the transfer characteristics, a plot point is defined by $I_D = I_{DSS}/4 = 6\text{ mA}/4 = 1.5\text{ mA}$ and $V_{GS} = V_P/2 = -3\text{ V}/2 = -1.5\text{ V}$. Considering the level of $V_P$ and the fact that Shockley's equation defines a curve that rises more rapidly as $V_{GS}$ becomes more positive, a plot point will be defined at $V_{GS} = +1\text{ V}$. Substituting into Shockley's equation yields

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= 6\text{ mA}\left(1 - \frac{+1\text{ V}}{-3\text{ V}}\right)^2 = 6\text{ mA}\left(1 + \frac{1}{3}\right)^2 = 6\text{ mA}(1.778)$$

$$= 10.67\text{ mA}$$

The resulting transfer curve appears in Fig. 6.30. Proceeding as described for JFETs, we have:

$$\text{Eq. (6.15):} \quad V_G = \frac{10\text{ M}\Omega(18\text{ V})}{10\text{ M}\Omega + 110\text{ M}\Omega} = 1.5\text{ V}$$

$$\text{Eq. (6.16):} \quad V_{GS} = V_G - I_D R_S = 1.5\text{ V} - I_D(750\ \Omega)$$

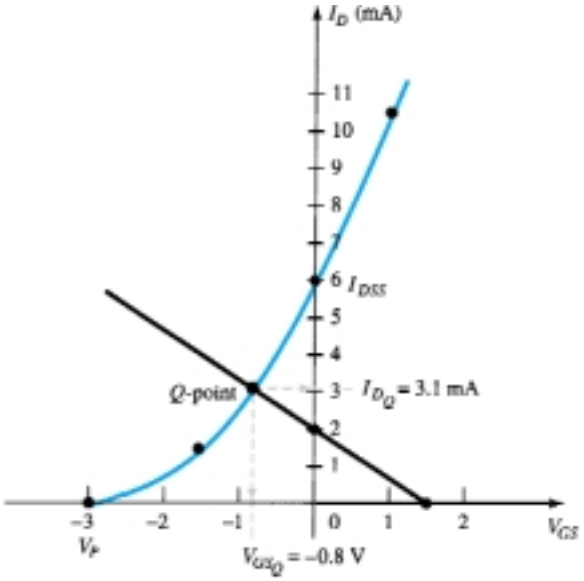



**Figure 6.30** Determining the *Q*-point for the network of Fig. 6.29.

Setting $I_D = 0\text{ mA}$ results in

$$V_{GS} = V_G = 1.5\text{ V}$$

Setting $V_{GS} = 0\text{ V}$ yields

$$I_D = \frac{V_G}{R_S} = \frac{1.5\text{ V}}{750\ \Omega} = 2\text{ mA}$$

The plot points and resulting bias line appear in Fig. 6.30. The resulting operating point:

$$I_{D_Q} = \mathbf{3.1\text{ mA}}$$

$$V_{GS_Q} = \mathbf{-0.8\text{ V}}$$

(b) Eq. (6.19): $V_{DS} = V_{DD} - I_D(R_D + R_S)$

$= 18\text{ V} - (3.1\text{ mA})(1.8\text{ k}\Omega + 750\ \Omega)$

$\cong \mathbf{10.1\text{ V}}$

## *EXAMPLE 6.8*

Repeat Example 6.7 with $R_S = 150\ \Omega$.

### Solution

(a) The plot points are the same for the transfer curve as shown in Fig. 6.31. For the bias line,

$$V_{GS} = V_G - I_D R_S = 1.5\text{ V} - I_D(150\ \Omega)$$

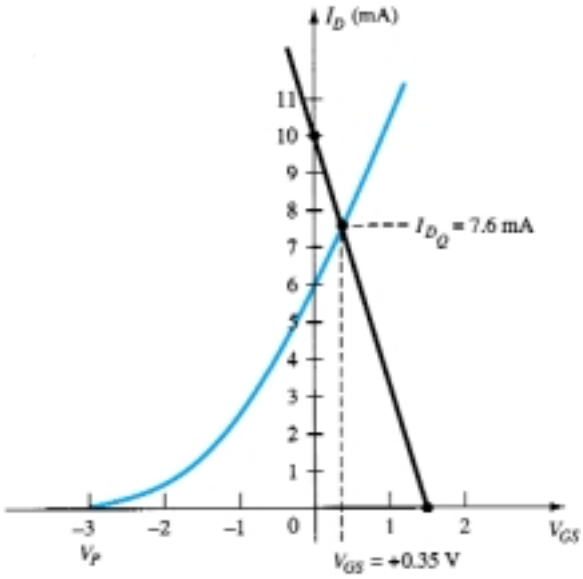



**Figure 6.31** Example 6.8.

Setting $I_D = 0$ mA results in

$$V_{GS} = 1.5\text{ V}$$

Setting $V_{GS} = 0$ V yields

$$I_D = \frac{V_G}{R_S} = \frac{1.5\text{ V}}{150\ \Omega} = 10\text{ mA}$$

The bias line is included on Fig. 6.31. Note in this case that the quiescent point results in a drain current that exceeds $I_{DSS}$, with a positive value for $V_{GS}$. The result:

$$I_{DQ} = \mathbf{7.6\text{ mA}}$$

$$V_{GSQ} = \mathbf{+0.35\text{ V}}$$

(b) Eq. (6.19): $V_{DS} = V_{DD} - I_D(R_D + R_S)$

$= 18\text{ V} - (7.6\text{ mA})(1.8\text{ k}\Omega + 150\ \Omega)$

$= \mathbf{3.18\text{ V}}$

## EXAMPLE 6.9

Determine the following for the network of Fig. 6.32.

(a) $I_{DQ}$ and $V_{GSQ}$.
(b) $V_D$.

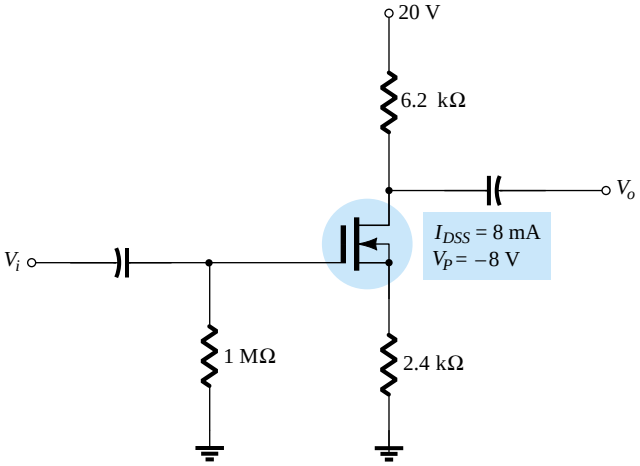



**Figure 6.32** Example 6.9.

### Solution

(a) The self-bias configuration results in

$$V_{GS} = -I_D R_S$$

as obtained for the JFET configuration, establishing the fact that $V_{GS}$ must be less than zero volts. There is therefore no requirement to plot the transfer curve for positive values of $V_{GS}$, although it was done on this occasion to complete the transfer characteristics. A plot point for the transfer characteristics for $V_{GS} < 0$ V is

$$I_D = \frac{I_{DSS}}{4} = \frac{8 \text{ mA}}{4} = 2 \text{ mA}$$

and

$$V_{GS} = \frac{V_P}{2} = \frac{-8 \text{ V}}{2} = -4 \text{ V}$$

and for $V_{GS} > 0$ V, since $V_P = -8$ V, we will choose

$$V_{GS} = +2 \text{ V}$$

and

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 8 \text{ mA}\left(1 - \frac{+2 \text{ V}}{-8 \text{ V}}\right)^2$$

$$= 12.5 \text{ mA}$$

The resulting transfer curve appears in Fig. 6.33. For the network bias line, at $V_{GS} = 0$ V, $I_D = 0$ mA. Choosing $V_{GS} = -6$ V gives

$$I_D = -\frac{V_{GS}}{R_S} = -\frac{-6 \text{ V}}{2.4 \text{ k}\Omega} = 2.5 \text{ mA}$$

The resulting $Q$-point:

$$I_{D_Q} = \mathbf{1.7 \text{ mA}}$$
$$V_{GSQ} = \mathbf{-4.3 \text{ V}}$$

(b) $V_D = V_{DD} - I_D R_D$
$= 20 \text{ V} - (1.7 \text{ mA})(6.2 \text{ k}\Omega)$
$= \mathbf{9.46 \text{ V}}$

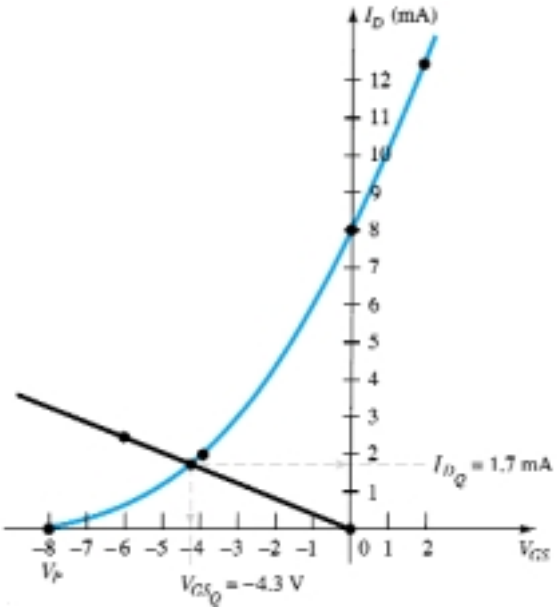



**Figure 6.33** Determining the *Q*-point for the network of Fig. 6.32.

The example to follow employs a design that can also be applied to JFET transistors. At first impression it appears rather simplistic, but in fact it often causes some confusion when first analyzed due to the special point of operation.

*EXAMPLE 6.10*

Determine $V_{DS}$ for the network of Fig. 6.34.

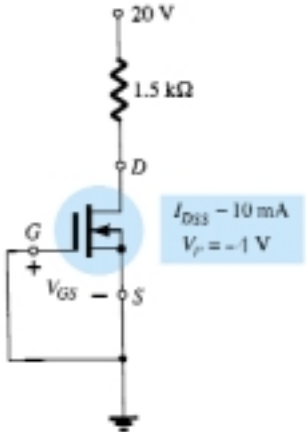



**Figure 6.34** Example 6.10.

### Solution

The direct connection between the gate and source terminals requires that

$$V_{GS} = 0 \text{ V}$$

Since $V_{GS}$ is fixed at 0 V, the drain current must be $I_{DSS}$ (by definition). In other words,

$$V_{GS_Q} = \mathbf{0 \text{ V}}$$

and

$$I_{D_Q} = \mathbf{10 \text{ mA}}$$

There is therefore no need to draw the transfer curve and

$$V_D = V_{DD} - I_D R_D = 20 \text{ V} - (10 \text{ mA})(1.5 \text{ k}\Omega)$$
$$= 20 \text{ V} - 15 \text{ V}$$
$$= \mathbf{5 \text{ V}}$$

## 6.6 ENHANCEMENT-TYPE MOSFETs

The transfer characteristics of the enhancement-type MOSFET are quite different from those encountered for the JFET and depletion-type MOSFETs, resulting in a graphical solution quite different from the preceding sections. First and foremost, recall that for the *n*-channel enhancement-type MOSFET, the drain current is zero for levels of gate-to-source voltage less than the threshold level $V_{GS(\text{Th})}$, as shown in Fig. 6.35. For levels of $V_{GS}$ greater than $V_{GS(\text{Th})}$, the drain current is defined by

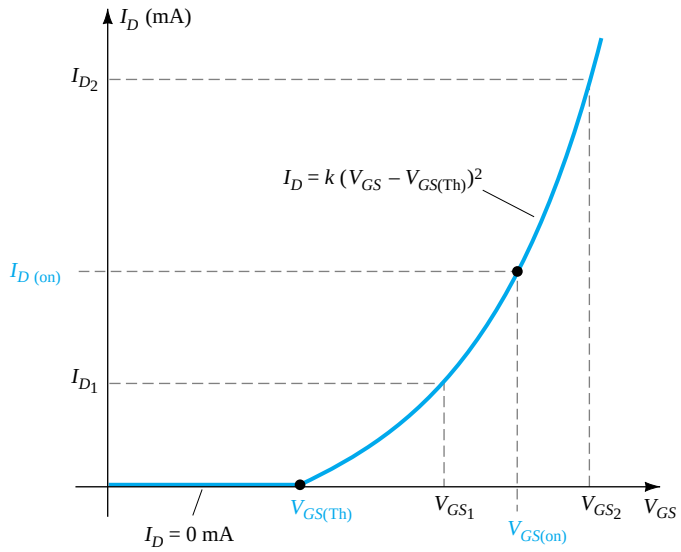



**Figure 6.35** Transfer characteristics of an *n*-channel enhancement-type MOSFET.

$$I_D = k(V_{GS} - V_{GS(\text{Th})})^2 \tag{6.25}$$

Since specification sheets typically provide the threshold voltage and a level of drain current ($I_{D(\text{on})}$) and its corresponding level of $V_{GS(\text{on})}$, two points are defined immediately as shown in Fig. 6.35. To complete the curve, the constant $k$ of Eq. (6.25) must be determined from the specification sheet data by substituting into Eq. (6.25) and solving for $k$ as follows:

$$I_D = k(V_{GS} - V_{GS(\text{Th})})^2$$

$$I_{D(\text{on})} = k(V_{GS(\text{on})} - V_{GS(\text{Th})})^2$$

and

$$k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_{GS(\text{Th})})^2} \tag{6.26}$$

Once $k$ is defined, other levels of $I_D$ can be determined for chosen values of $V_{GS}$. Typically, a point between $V_{GS(\text{Th})}$ and $V_{GS(\text{on})}$ and one just greater than $V_{GS(\text{on})}$ will provide a sufficient number of points to plot Eq. (6.25) (note $I_{D_1}$ and $I_{D_2}$ on Fig. 6.35).

## Feedback Biasing Arrangement

A popular biasing arrangement for enhancement-type MOSFETs is provided in Fig. 6.36. The resistor $R_G$ brings a suitably large voltage to the gate to drive the MOSFET "on." Since $I_G = 0$ mA and $V_{R_G} = 0$ V, the dc equivalent network appears as shown in Fig. 6.37.

A direct connection now exists between drain and gate, resulting in

$$V_D = V_G$$

and

$$V_{DS} = V_{GS} \tag{6.27}$$

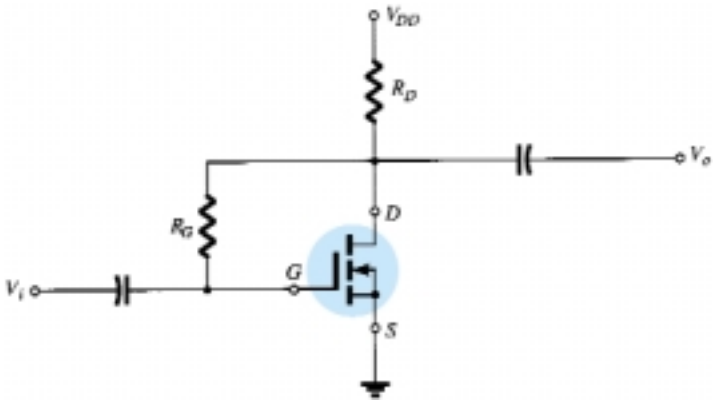



**Figure 6.36** Feedback biasing arrangement.

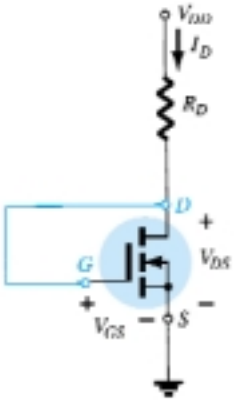



**Figure 6.37** DC equivalent of the network of Fig. 6.36.

For the output circuit,

$$V_{DS} = V_{DD} - I_D R_D$$

which becomes the following after substituting Eq. (6.27):

$$\boxed{V_{GS} = V_{DD} - I_D R_D} \tag{6.28}$$

The result is an equation that relates the same two variables as Eq. (6.25), permitting the plot of each on the same set of axes.

Since Eq. (6.28) is that of a straight line, the same procedure described earlier can be employed to determine the two points that will define the plot on the graph. Substituting $I_D = 0$ mA into Eq. (6.28) gives

$$\boxed{V_{GS} = V_{DD}|_{I_D = 0\text{ mA}}} \tag{6.29}$$

Substituting $V_{GS} = 0$ V into Eq. (6.28), we have

$$\boxed{I_D = \left.\frac{V_{DD}}{R_D}\right|_{V_{GS} = 0\text{ V}}} \tag{6.30}$$

The plots defined by Eqs. (6.25) and (6.28) appear in Fig. 6.38 with the resulting operating point.

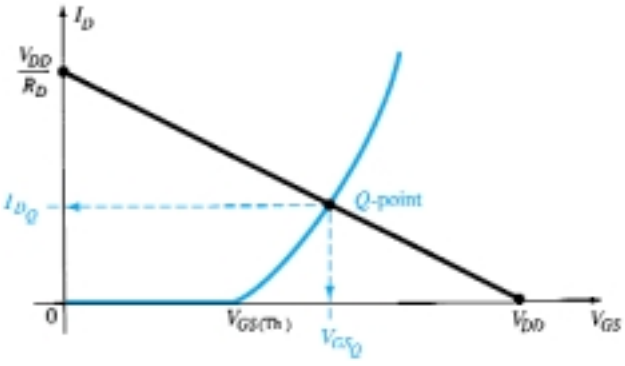



**Figure 6.38** Determining the *Q*-point for the network of Fig. 6.36.

Determine $I_{D_Q}$ and $V_{DS_Q}$ for the enhancement-type MOSFET of Fig. 6.39. *EXAMPLE 6.11*

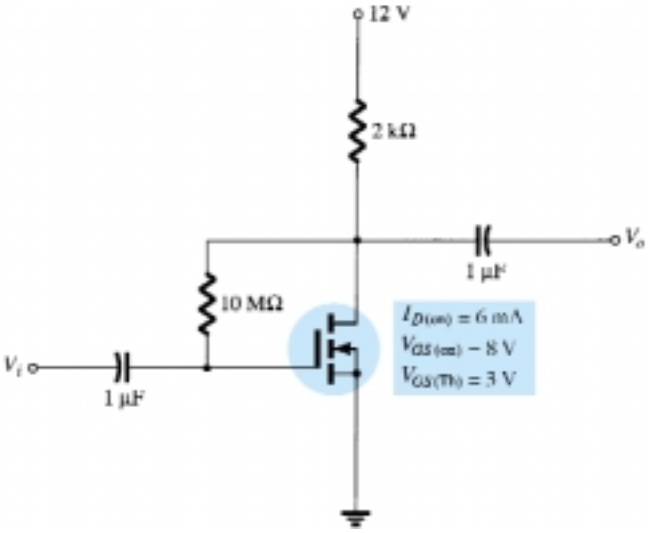



**Figure 6.39** Example 6.11.

## Solution

**Plotting the Transfer Curve:**

Two points are defined immediately as shown in Fig. 6.40. Solving for $k$:

$$\text{Eq. (6.26):} \quad k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_{GS(\text{Th})})^2}$$

$$= \frac{6 \text{ mA}}{(8 \text{ V} - 3 \text{ V})^2} = \frac{6 \times 10^{-3}}{25} \text{A/v}^2$$

$$= \mathbf{0.24 \times 10^{-3} \text{ A/V}^2}$$

For $V_{GS} = 6$ V (between 3 and 8 V):

$$I_D = 0.24 \times 10^{-3}(6 \text{ V} - 3 \text{ V})^2 = 0.24 \times 10^{-3}(9)$$

$$= 2.16 \text{ mA}$$

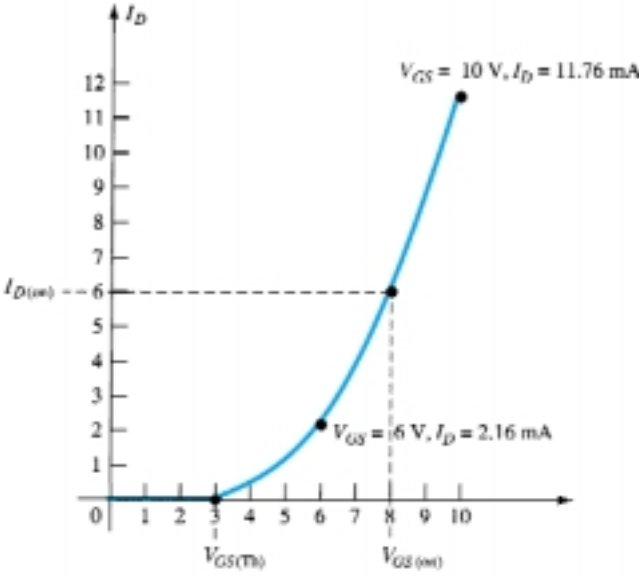



**Figure 6.40** Plotting the transfer curve for the MOSFET of Fig. 6.39.

as shown on Fig. 6.40. For $V_{GS} = 10$ V (slightly greater than $V_{GS(Th)}$):

$$I_D = 0.24 \times 10^{-3}(10\text{ V} - 3\text{ V})^2 = 0.24 \times 10^{-3}(49)$$
$$= 11.76\text{ mA}$$

as also appearing on Fig. 6.40. The four points are sufficient to plot the full curve for the range of interest as shown in Fig. 6.40.

**For the Network Bias Line:**

$$V_{GS} = V_{DD} - I_D R_D$$
$$= 12\text{ V} - I_D(2\text{ k}\Omega)$$

Eq. (6.29): $V_{GS} = V_{DD} = 12\text{ V}\big|_{I_D = 0\text{ mA}}$

Eq. (6.30): $I_D = \dfrac{V_{DD}}{R_D} = \dfrac{12\text{ V}}{2\text{ k}\Omega} = 6\text{ mA}\big|_{V_{GS} = 0\text{ V}}$

The resulting bias line appears in Fig. 6.41.

At the operating point:

$$I_{D_Q} = \mathbf{2.75\text{ mA}}$$

and $$V_{GS_Q} = 6.4\text{ V}$$

with $$V_{DS_Q} = V_{GS_Q} = \mathbf{6.4\text{ V}}$$

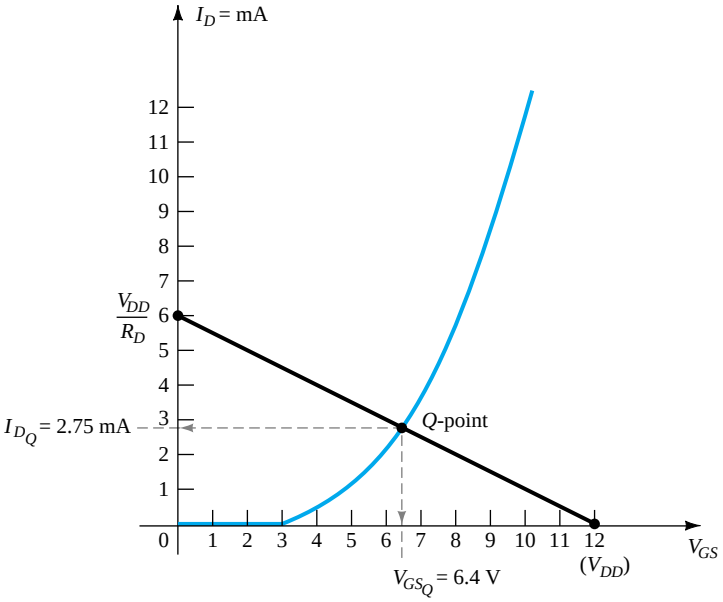



**Figure 6.41** Determining the *Q*-point for the network of Fig. 6.39.

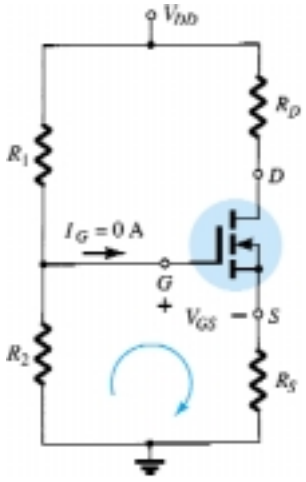



**Figure 6.42** Voltage-divider biasing arrangement for an *n*-channel enhancement MOSFET.

## Voltage-Divider Biasing Arrangement

A second popular biasing arrangement for the enhancement-type MOSFET appears in Fig. 6.42. The fact that $I_G = 0$ mA results in the following equation for $V_{GG}$ as derived from an application of the voltage-divider rule:

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} \tag{6.31}$$

Applying Kirchhoff's voltage law around the indicated loop of Fig. 6.42 will result in

$$+V_G - V_{GS} - V_{R_S} = 0$$

and

$$V_{GS} = V_G - V_{R_S}$$

or

$$\boxed{V_{GS} = V_G - I_D R_S} \tag{6.32}$$

For the output section:

$$V_{R_S} + V_{DS} + V_{R_D} - V_{DD} = 0$$

and

$$V_{DS} = V_{DD} - V_{R_S} - V_{R_D}$$

or

$$\boxed{V_{DS} = V_{DD} - I_D(R_S + R_D)} \tag{6.33}$$

Since the characteristics are a plot of $I_D$ versus $V_{GS}$ and Eq. (6.32) relates the same two variables, the two curves can be plotted on the same graph and a solution determined at their intersection. Once $I_{D_Q}$ and $V_{GS_Q}$ are known, all the remaining quantities of the network such as $V_{DS}$, $V_D$, and $V_S$ can be determined.

---

*EXAMPLE 6.12*

Determine $I_{D_Q}$, $V_{GS_Q}$, and $V_{DS}$ for the network of Fig. 6.43.

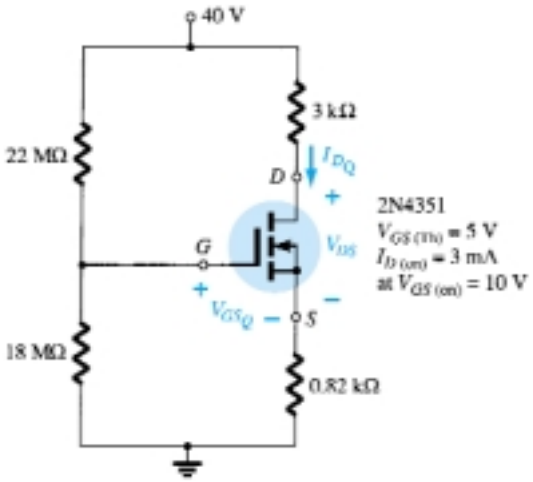



**Figure 6.43** Example 6.12.

## Solution

**Network:**

$$\text{Eq. (6.31):} \quad V_G = \frac{R_2 V_{DD}}{R_1 + R_2} = \frac{(18\ \text{M}\Omega)(40\ \text{V})}{22\ \text{M}\Omega + 18\ \text{M}\Omega} = 18\ \text{V}$$

$$\text{Eq. (6.32):} \quad V_{GS} = V_G - I_D R_S = 18\ \text{V} - I_D(0.82\ \text{k}\Omega)$$

When $I_D = 0$ mA,

$$V_{GS} = 18\ \text{V} - (0\ \text{mA})(0.82\ \text{k}\Omega) = 18\ \text{V}$$

as appearing on Fig. 6.44. When $V_{GS} = 0$ V,

$$V_{GS} = 18\ \text{V} - I_D(0.82\ \text{k}\Omega)$$

$$0 = 18\ \text{V} - I_D(0.82\ \text{k}\Omega)$$

$$I_D = \frac{18\ \text{V}}{0.82\ \text{k}\Omega} = 21.95\ \text{mA}$$

as appearing on Fig. 6.44.

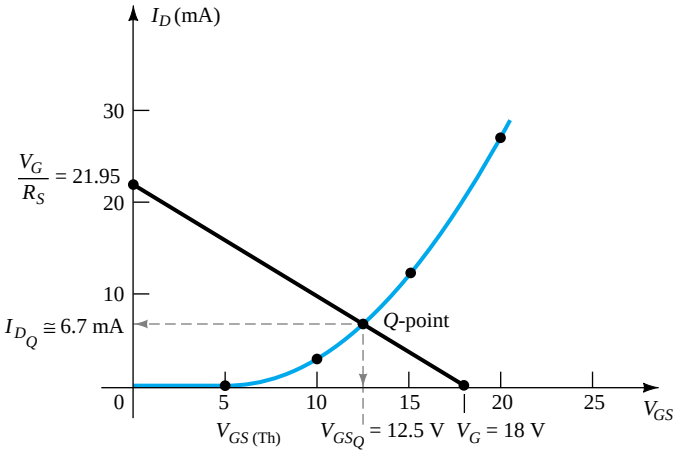



**Figure 6.44** Determining the *Q*-point for the network of Example 6.12.

**Device:**

$$V_{GS(\text{Th})} = 5\text{ V}, \qquad I_{D(\text{on})} = 3\text{ mA with } V_{GS(\text{on})} = 10\text{ V}$$

$$\text{Eq. (6.26):} \quad k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_{GS(\text{Th})})^2}$$

$$= \frac{3\text{ mA}}{(10\text{ V} - 5\text{ V})^2} = 0.12 \times 10^{-3}\text{ A/V}^2$$

and

$$I_D = k(V_{GS} - V_{GS(\text{Th})})^2$$

$$= 0.12 \times 10^{-3}(V_{GS} - 5)^2$$

which is plotted on the same graph (Fig. 6.44). From Fig. 6.44,

$$I_{D_Q} \cong \mathbf{6.7\text{ mA}}$$

$$V_{GS_Q} = \mathbf{12.5\text{ V}}$$

$$\text{Eq. (6.33):} \quad V_{DS} = V_{DD} - I_D(R_S + R_D)$$

$$= 40\text{ V} - (6.7\text{ mA})(0.82\text{ k}\Omega + 3.0\text{ k}\Omega)$$

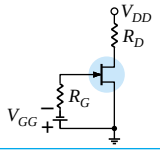
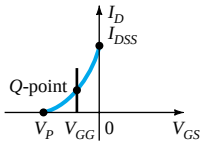
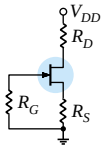
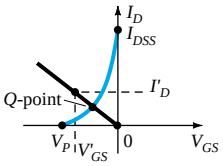
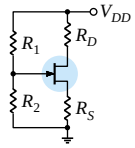
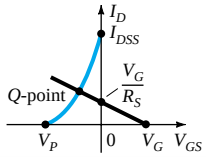
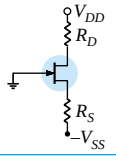
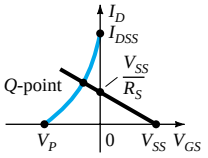
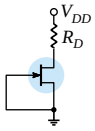
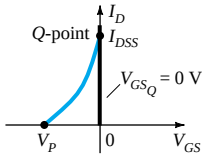
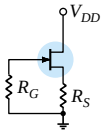
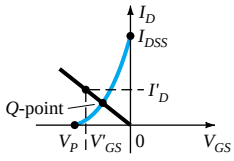
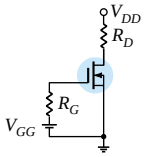
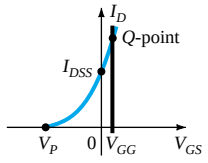
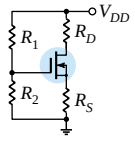
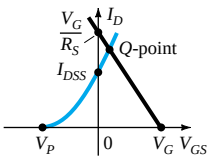
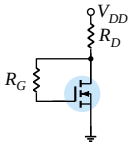
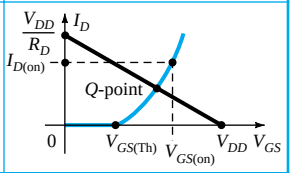
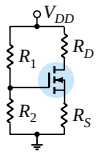
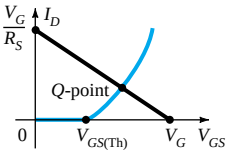
$$= 40\text{ V} - 25.6\text{ V}$$

$$= \mathbf{14.4\text{ V}}$$

## 6.7 SUMMARY TABLE

Now that the most popular biasing arrangements for the various FETs have been introduced, Table 6.1 reviews the basic results and demonstrates the similarity in approach for a number of configurations. It also reveals that the general analysis of dc configurations for FETs is not overly complex. Once the transfer characteristics are established, the network self-bias line can be drawn and the *Q*-point determined at the intersection of the device transfer characteristic and the network bias curve. The remaining analysis is simply an application of the basic laws of circuit analysis.

**TABLE 6.1 FET Bias Configurations**

| *Type* | *Configuration* | *Pertinent Equations* | *Graphical Solution* |
|---|---|---|---|
| JFET<br>Fixed-bias |  | $V_{GSQ} = -V_{GG}$<br>$V_{DS} = V_{DD} - I_D R_S$ |  |
| JFET<br>Self-bias |  | $V_{GS} = -I_D R_S$<br>$V_{DS} = V_{DD} - I_D(R_D + R_S)$ |  |
| JFET<br>Voltage-divider<br>bias |  | $V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$<br>$V_{GS} = V_G - I_D R_S$<br>$V_{DS} = V_{DD} - I_D(R_D + R_S)$ |  |
| JFET<br>Common-gate |  | $V_{GS} = V_{SS} - I_D R_S$<br>$V_{DS} = V_{DD} + V_{SS} - I_D(R_D + R_S)$ |  |
| JFET<br>($V_{GSQ} = 0$ V) |  | $V_{GSQ} = 0$ V<br>$I_{DQ} = I_{DSS}$ |  |
| JFET<br>($R_D = 0\ \Omega$) |  | $V_{GS} = -I_D R_S$<br>$V_D = V_{DD}$<br>$V_S = I_D R_S$<br>$V_{DS} = V_{DD} - I_S R_S$ |  |
| Depletion-type<br>MOSFET<br>Fixed-bias |  | $V_{GSQ} = +V_{GG}$<br>$V_{DS} = V_{DD} - I_D R_S$ |  |
| Depletion-type<br>MOSFET<br>Voltage-divider<br>bias |  | $V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$<br>$V_{GS} = V_G - I_S R_S$<br>$V_{DS} = V_{DD} - I_D(R_D + R_S)$ |  |
| Enhancement<br>type MOSFET<br>Feedback<br>configuration |  | $V_{GS} = V_{DS}$<br>$V_{GS} = V_{DD} - I_D R_D$ |  |
| Enhancement-<br>type MOSFET<br>Voltage-divider<br>bias |  | $V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$<br>$V_{GS} = V_G - I_D R_S$ |  |

## 6.8 COMBINATION NETWORKS

Now that the dc analysis of a variety of BJT and FET configurations is established, the opportunity to analyze networks with both types of devices presents itself. Fundamentally, the analysis simply requires that we *first* approach the device that will provide a terminal voltage or current level. The door is then usually open to calculate other quantities and concentrate on the remaining unknowns. These are usually particularly interesting problems due to the challenge of finding the opening and then using the results of the past few sections and Chapter 5 to find the important quantities for each device. The equations and relationships used are simply those we have now employed on more than one occasion—no need to develop any new methods of analysis.

*EXAMPLE 6.13* Determine the levels of $V_D$ and $V_C$ for the network of Fig. 6.45.

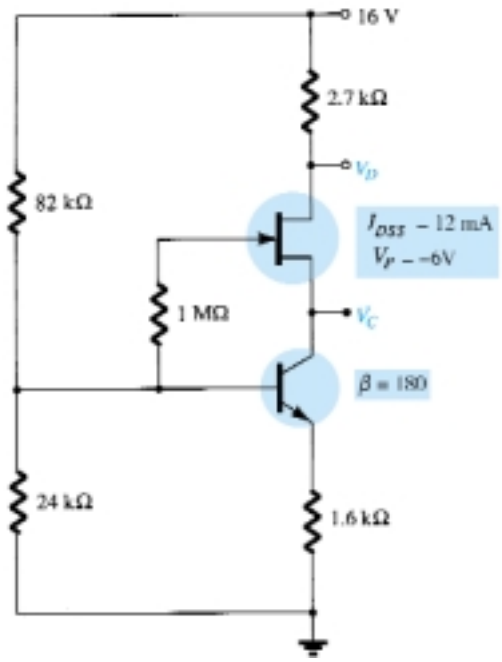



**Figure 6.45** Example 6.13.

### Solution

From past experience we now realize that $V_{GS}$ is typically an important quantity to determine or write an equation for when analyzing JFET networks. Since $V_{GS}$ is a level for which an immediate solution is not obvious, let us turn our attention to the transistor configuration. The voltage-divider configuration is one where the approximate technique can be applied ($\beta R_E = (180 \times 1.6\,\text{k}\Omega) = 288\,\text{k}\Omega > 10R_2 = 240\,\text{k}\Omega$), permitting a determination of $V_B$ using the voltage-divider rule on the input circuit.

For $V_B$:

$$V_B = \frac{24\,\text{k}\Omega(16\text{ V})}{82\,\text{k}\Omega + 24\,\text{k}\Omega} = 3.62\text{ V}$$

Using the fact that $V_{BE} = 0.7$ V results in

$$V_E = V_B - V_{BE} = 3.62\text{ V} - 0.7\text{ V}$$
$$= 2.92\text{ V}$$

and $$I_E = \frac{V_{RE}}{R_E} = \frac{V_E}{R_E} = \frac{2.92\ \text{V}}{1.6\ \text{k}\Omega} = 1.825\ \text{mA}$$

with $$I_C \cong I_E = 1.825\ \text{mA}$$

Continuing, we find for this configuration that

$$I_D = I_S = I_C$$

and $$V_D = 16\ \text{V} - I_D(2.7\ \text{k}\Omega)$$
$$= 16\ \text{V} - (1.825\ \text{mA})(2.7\ \text{k}\Omega) = 16\ \text{V} - 4.93\ \text{V}$$
$$= \mathbf{11.07\ V}$$

The question of how to determine $V_C$ is not as obvious. Both $V_{CE}$ and $V_{DS}$ are unknown quantities preventing us from establishing a link between $V_D$ and $V_C$ or from $V_E$ to $V_D$. A more careful examination of Fig. 6.45 reveals that $V_C$ is linked to $V_B$ by $V_{GS}$ (assuming that $V_{R_G} = 0$ V). Since we know $V_B$ if we can find $V_{GS}$, $V_C$ can be determined from

$$V_C = V_B - V_{GS}$$

The question then arises as to how to find the level of $V_{GS_Q}$ from the quiescent value of $I_D$. The two are related by Shockley's equation:

$$I_{D_Q} = I_{DSS}\left(1 - \frac{V_{GS_Q}}{V_P}\right)^2$$

and $V_{GS_Q}$ could be found mathematically by solving for $V_{GS_Q}$ and substituting numerical values. However, let us turn to the graphical approach and simply work in the reverse order employed in the preceding sections. The JFET transfer characteristics are first sketched as shown in Fig. 6.46. The level of $I_{D_Q}$ is then established by a horizontal line as shown in the same figure. $V_{GS_Q}$ is then determined by dropping a line down from the operating point to the horizontal axis, resulting in

$$V_{GS_Q} = \mathbf{-3.7\ V}$$

The level of $V_C$:

$$V_C = V_B - V_{GS_Q} = 3.62\ \text{V} - (-3.7\ \text{V})$$
$$= \mathbf{7.32\ V}$$

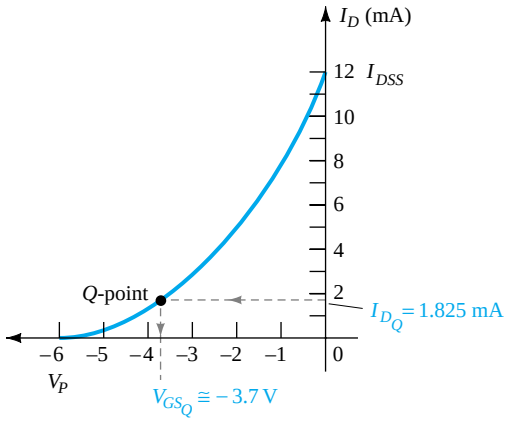



**Figure 6.46** Determining the *Q*-point for the network of Fig. 6.45.

*EXAMPLE 6.14* Determine $V_D$ for the network of Fig. 6.47.

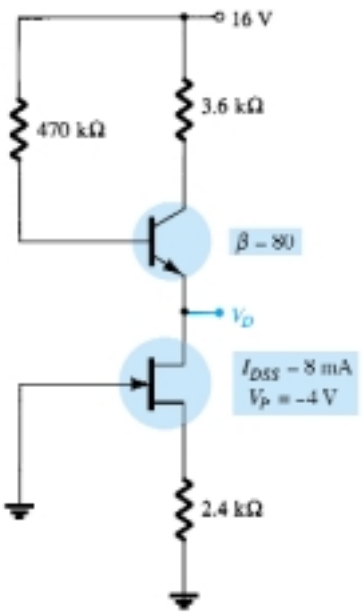



**Figure 6.47** Example 6.14.

## Solution

In this case, there is no obvious path to determine a voltage or current level for the transistor configuration. However, turning to the self-biased JFET, an equation for $V_{GS}$ can be derived and the resulting quiescent point determined using graphical techniques. That is,

$$V_{GS} = -I_D R_S = -I_D(2.4\ \text{k}\Omega)$$

resulting in the self-bias line appearing in Fig. 6.48 that establishes a quiescent point at

$$V_{GS_Q} = -2.6\ \text{V}$$
$$I_{D_Q} = 1\ \text{mA}$$

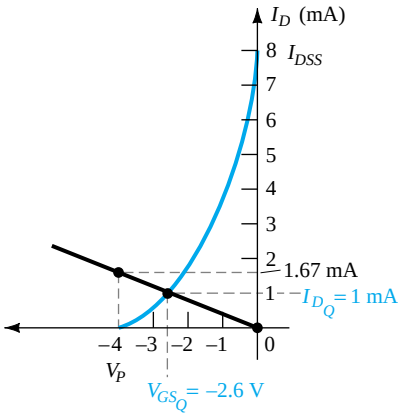



**Figure 6.48** Determining the *Q*-point for the network of Fig. 6.47.

For the transistor,

$$I_E \cong I_C = I_D = 1\ \text{mA}$$

and

$$I_B = \frac{I_C}{\beta} = \frac{1\ \text{mA}}{80} = 12.5\ \mu\text{A}$$

$$V_B = 16\ \text{V} - I_B(470\ \text{k}\Omega)$$
$$= 16\ \text{V} - (12.5\ \mu\text{A})(470\ \text{k}\Omega) = 16\ \text{V} - 5.875\ \text{V}$$
$$= 10.125\ \text{V}$$

and

$$V_E = V_D = V_B - V_{BE}$$
$$= 10.125\ \text{V} - 0.7\ \text{V}$$
$$= \mathbf{9.425\ V}$$

## 6.9 DESIGN

The design process is one that is not limited solely to dc conditions. The area of application, level of amplification desired, signal strength, and operating conditions are just a few of the conditions that enter into the total design process. However, we will first concentrate on establishing the chosen dc conditions.

For example, if the levels of $V_D$ and $I_D$ are specified for the network of Fig. 6.49, the level of $V_{GS_Q}$ can be determined from a plot of the transfer curve and $R_S$ can then be determined from $V_{GS} = -I_D R_S$. If $V_{DD}$ is specified, the level of $R_D$ can then be calculated from $R_D = (V_{DD} - V_D)/I_D$. Of course, the value of $R_S$ and $R_D$ may not be standard commercial values, requiring that the nearest commercial value be employed. However, with the tolerance (range of values) normally specified for the parameters of a network, the slight variation due to the choice of standard values will seldom cause a real concern in the design process.

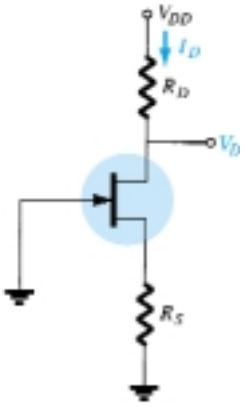



**Figure 6.49** Self-bias configuration to be designed.

The above is only one possibility for the design phase involving the network of Fig. 6.49. It is possible that only $V_{DD}$ and $R_D$ are specified together with the level of $V_{DS}$. The device to be employed may have to be specified along with the level of $R_S$. It appears logical that the device chosen should have a maximum $V_{DS}$ greater than the specified value by a safe margin.

In general, it is good design practice for linear amplifiers to choose operating points that do not crowd the saturation level ($I_{DSS}$) or cutoff ($V_P$) regions. Levels of $V_{GS_Q}$ close to $V_P/2$ or $I_{D_Q}$ near $I_{DSS}/2$ are certainly reasonable starting points in the design. Of course, in every design procedure the maximum levels of $I_D$ and $V_{DS}$ as appearing on the specification sheet must not be considered as exceeded.

The examples to follow have a design or synthesis orientation in that specific levels are provided and network parameters such as $R_D$, $R_S$, $V_{DD}$, and so on, must be determined. In any case, the approach is in many ways the opposite of that described in previous sections. In some cases, it is just a matter of applying Ohm's law in its appropriate form. In particular, if resistive levels are requested, the result is often obtained simply by applying Ohm's law in the following form:

$$R_{\text{unknown}} = \frac{V_R}{I_R} \tag{6.34}$$

where $V_R$ and $I_R$ are often parameters that can be found directly from the specified voltage and current levels.

---

*EXAMPLE 6.15*

For the network of Fig. 6.50, the levels of $V_{D_Q}$ and $I_{D_Q}$ are specified. Determine the required values of $R_D$ and $R_S$. What are the closest standard commercial values?

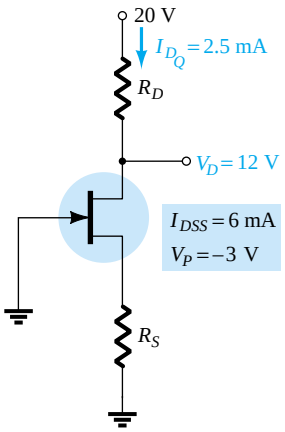



**Figure 6.50** Example 6.15.

## Solution

As defined by Eq. (6.34),

$$R_D = \frac{V_{R_D}}{I_{D_Q}} = \frac{V_{DD} - V_{D_Q}}{I_{D_Q}}$$

and

$$= \frac{20\text{ V} - 12\text{ V}}{2.5\text{ mA}} = \frac{8\text{ V}}{2.5\text{ mA}} = \mathbf{3.2\ k\Omega}$$

Plotting the transfer curve in Fig. 6.51 and drawing a horizontal line at $I_{D_Q} = 2.5$ mA will result in $V_{GS_Q} = -1$ V, and applying $V_{GS} = -I_D R_S$ will establish the level of $R_S$:

$$R_S = \frac{-(V_{GS_Q})}{I_{D_Q}} = \frac{-(-1\text{ V})}{2.5\text{ mA}} = \mathbf{0.4\ k\Omega}$$

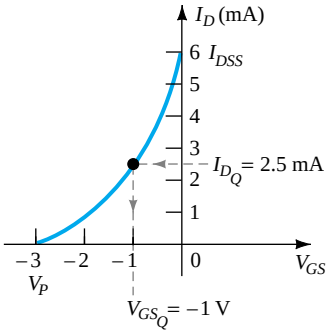



**Figure 6.51** Determining $V_{GS_Q}$ for the network of Fig. 6.50.

The nearest standard commercial values are

$$R_D = 3.2\text{ k}\Omega \Rightarrow \mathbf{3.3\ k\Omega}$$

$$R_S = 0.4\text{ k}\Omega \Rightarrow \mathbf{0.39\ k\Omega}$$

## *EXAMPLE 6.16*

For the voltage-divider bias configuration of Fig. 6.52, if $V_D = 12$ V and $V_{GS_Q} = -2$ V, determine the value of $R_S$.

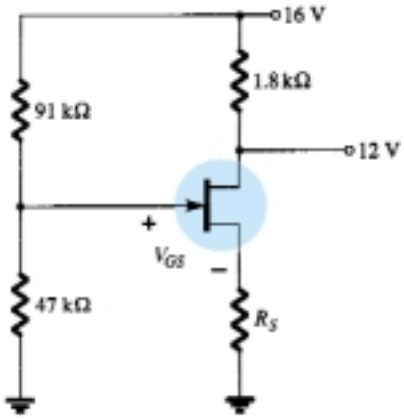



**Figure 6.52** Example 6.16.

## Solution

The level of $V_G$ is determined as follows:

$$V_G = \frac{47\text{ k}\Omega(16\text{ V})}{47\text{ k}\Omega + 91\text{ k}\Omega} = 5.44\text{ V}$$

with

$$I_D = \frac{V_{DD} - V_D}{R_D}$$

$$= \frac{16\text{ V} - 12\text{ V}}{1.8\text{ k}\Omega} = 2.22\text{ mA}$$

The equation for $V_{GS}$ is then written and the known values substituted:

$$V_{GS} = V_G - I_D R_S$$

$$-2\text{ V} = 5.44\text{ V} - (2.22\text{ mA})R_S$$

$$-7.44\text{ V} = -(2.22\text{ mA})R_S$$

and

$$R_S = \frac{7.44\text{ V}}{2.22\text{ mA}} = \mathbf{3.35\ k\Omega}$$

The nearest standard commercial value is 3.3 kΩ.

## EXAMPLE 6.17

The levels of $V_{DS}$ and $I_D$ are specified as $V_{DS} = \frac{1}{2}V_{DD}$ and $I_D = I_{D(on)}$ for the network of Fig. 6.53. Determine the level of $V_{DD}$ and $R_D$.

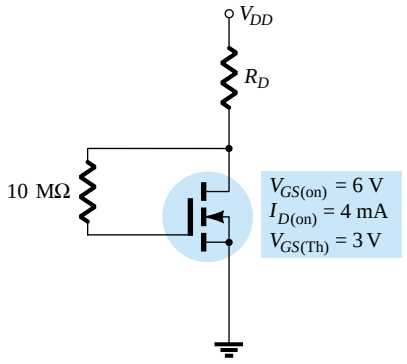



**Figure 6.53** Example 6.17.

### Solution

Given $I_D = I_{D(on)} = 4$ mA and $V_{GS} = V_{GS(on)} = 6$ V, for this configuration,

$$V_{DS} = V_{GS} = \tfrac{1}{2}V_{DD}$$

and

$$6 \text{ V} = \tfrac{1}{2}V_{DD}$$

so that

$$V_{DD} = \mathbf{12\ V}$$

Applying Eq. (6.34) yields

$$R_D = \frac{V_{R_D}}{I_D} = \frac{V_{DD} - V_{DS}}{I_{D(on)}} = \frac{V_{DD} - \frac{1}{2}V_{DD}}{I_{D(on)}} = \frac{\frac{1}{2}V_{DD}}{I_{D(on)}}$$

and

$$R_D = \frac{6 \text{ V}}{4 \text{ mA}} = \mathbf{1.5\ k\Omega}$$

which is a standard commercial value.

## 6.10 TROUBLESHOOTING

How often has a network been carefully constructed only to find that when the power is applied, the response is totally unexpected and fails to match the theoretical calculations. What is the next step? Is it a bad connection? A misreading of the color code for a resistive element? An error in the construction process? The range of possibilities seems vast and often frustrating. The troubleshooting process first described in the analysis of BJT transistor configurations should narrow down the list of possibilities and isolate the problem area following a definite plan of attack. In general, the process begins with a rechecking of the network construction and the terminal connections. This is usually followed by the checking of voltage levels between specific terminals and ground or between terminals of the network. Seldom are current levels measured since such maneuvers require disturbing the network structure to insert the meter. Of course, once the voltage levels are obtained, current levels can be calculated using Ohm's law. In any case, some idea of the expected voltage or current level must be known for the measurement to have any importance. In total, therefore, the troubleshooting process can begin with some hope of success only if the basic operation of the network is understood along with some expected levels of voltage

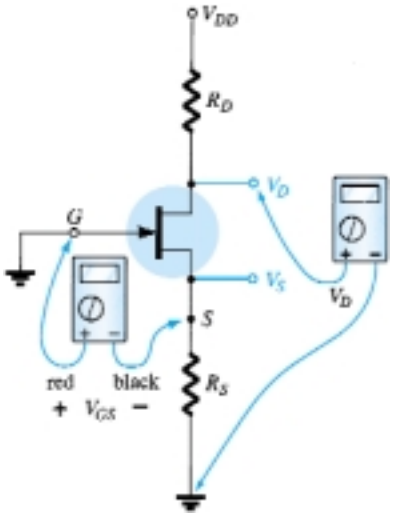



**Figure 6.54** Checking the dc operation of the JFET self-bias configuration.

or current. For the *n*-channel JFET amplifier, it is clearly understood that the quiescent value of $V_{GS_Q}$ is limited to 0 V or a negative voltage. For the network of Fig. 6.54, $V_{GS_Q}$ is limited to negative values in the range 0 V to $V_P$. If a meter is hooked up as shown in Fig. 6.54, with the positive lead (normally red) to the gate and the negative lead (usually black) to the source, the resulting reading should have a negative sign and a magnitude of a few volts. Any other response should be considered suspicious and needs to be investigated.

The level of $V_{DS}$ is typically between 25% and 75% of $V_{DD}$. A reading of 0 V for $V_{DS}$ clearly indicates that either the output circuit has an "open" or the JFET is internally short-circuited between drain and source. If $V_D$ is $V_{DD}$ volts, there is obviously no drop across $R_D$ due to the lack of current through $R_D$ and the connections should be checked for continuity.

If the level of $V_{DS}$ seems inappropriate, the continuity of the output circuit can easily be checked by grounding the negative lead of the voltmeter and measuring the voltage levels from $V_{DD}$ to ground using the positive lead. If $V_D = V_{DD}$, the current through $R_D$ may be zero, but there is continuity between $V_D$ and $V_{DD}$. If $V_S = V_{DD}$, the device is not open between drain and source, but it is also not "on." The continuity through to $V_S$ is confirmed, however. In this case, it is possible that there is a poor ground connection between $R_S$ and ground that may not be obvious. The internal connection between the wire of your lead and the terminal connector may have separated. Other possibilities also exist, such as a shorted device from drain to source, but the troubleshooter will simply have to narrow down the possible causes for the malfunction.

The continuity of a network can also be checked simply by measuring the voltage across any resistor of the network (except for $R_G$ in the JFET configuration). An indication of 0 V immediately reveals the lack of current through the element due to an open circuit in the network.

The most sensitive element in the BJT and JFET configurations is the amplifier itself. The application of excessive voltage during the construction or testing phase or the use of incorrect resistor values resulting in high current levels can destroy the device. If you question the condition of the amplifier, the best test for the FET is the curve tracer since it not only reveals whether the device is operable but also its range of current and voltage levels. Some testers may reveal that the device is still fundamentally sound but do not reveal whether its range of operation has been severely reduced.

The development of good troubleshooting techniques comes primarily from experience and a level of confidence in what to expect and why. There are, of course, times when the reasons for a strange response seem to disappear mysteriously when you check a network. In such cases, it is best not to breathe a sigh of relief and continue with the construction. The cause for such a sensitive "make or break" situation should be found and corrected, or it may reoccur at the most inopportune moment.

## 6.11 *P*-CHANNEL FETS

The analysis thus far has been limited solely to *n*-channel FETs. For *p*-channel FETs, a mirror image of the transfer curves is employed, and the defined current directions are reversed as shown in Fig. 6.55 for the various types of FETs.

Note for each configuration of Fig. 6.55 that each supply voltage is now a negative voltage drawing current in the indicated direction. In particular, note that the double-subscript notation for voltages continues as defined for the *n*-channel device: $V_{GS}$, $V_{DS}$, and so on. In this case, however, $V_{GS}$ is positive (positive or negative for the depletion-type MOSFET) and $V_{DS}$ negative.

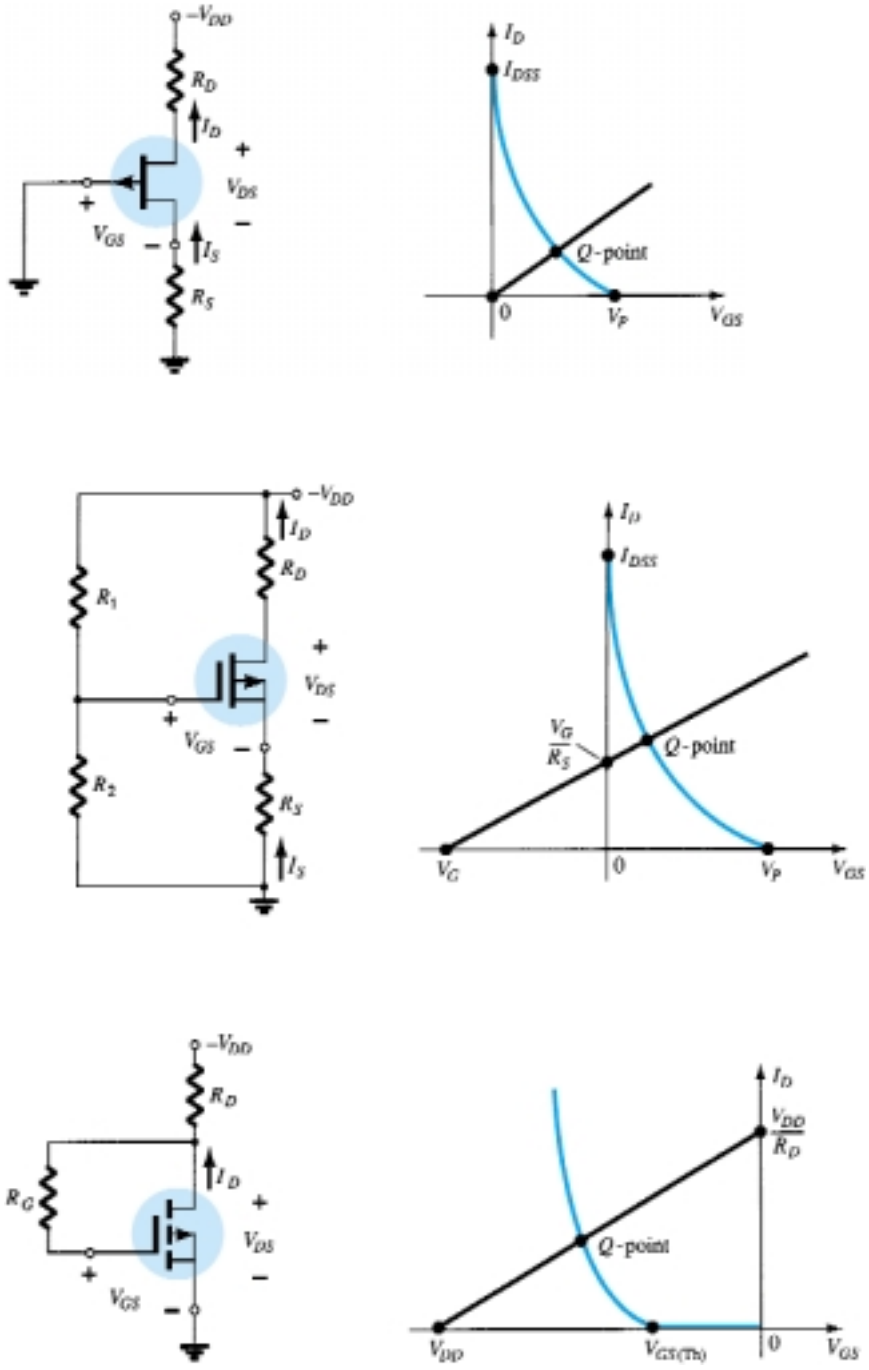



**Figure 6.55** *p*-channel configurations.

Due to the similarities between the analysis of *n*-channel and *p*-channel devices, one can actually assume an *n*-channel device and reverse the supply voltage and perform the entire analysis. When the results are obtained, the magnitude of each quantity will be correct, although the current direction and voltage polarities will have to be reversed. However, the next example will demonstrate that with the experience gained through the analysis of *n*-channel devices, the analysis of *p*-channel devices is quite straightforward.

*EXAMPLE 6.18* Determine $I_{DQ}$, $V_{GSQ}$, and $V_{DS}$ for the *p*-channel JFET of Fig. 6.56.

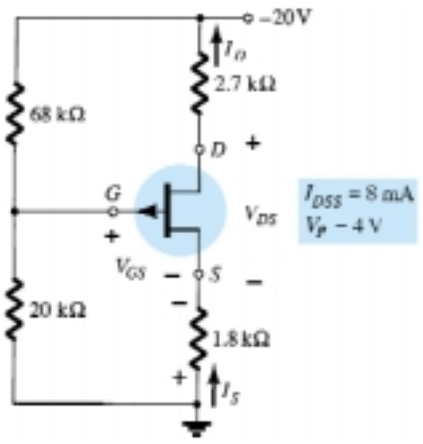



**Figure 6.56** Example 6.18.

## Solution

$$V_G = \frac{20\ \text{k}\Omega(-20\ \text{V})}{20\ \text{k}\Omega + 68\ \text{k}\Omega} = -4.55\ \text{V}$$

Applying Kirchhoff's voltage law gives

$$V_G - V_{GS} + I_D R_S = 0$$

and

$$V_{GS} = V_G + I_D R_S$$

Choosing $I_D = 0$ mA yields

$$V_{GS} = V_G = -4.55\ \text{V}$$

as appearing in Fig. 6.57.
Choosing $V_{GS} = 0$ V, we obtain

$$I_D = -\frac{V_G}{R_S} = -\frac{-4.55\ \text{V}}{1.8\ \text{k}\Omega} = 2.53\ \text{mA}$$

as also appearing in Fig. 6.57.
The resulting quiescent point from Fig. 6.57:

$$I_{DQ} = \mathbf{3.4\ mA}$$

$$V_{GSQ} = \mathbf{1.4\ V}$$

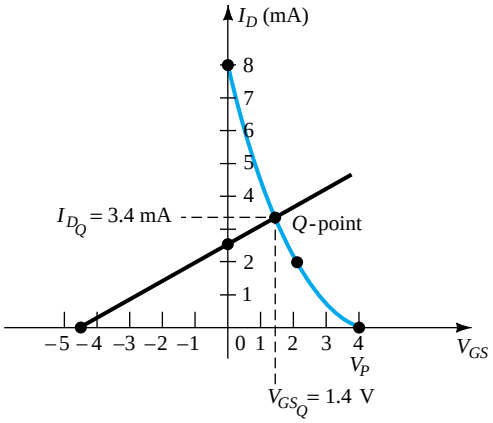



**Figure 6.57** Determining the *Q*-point for the JFET configuration of Fig. 6.56.

For $V_{DS}$, Kirchhoff's voltage law will result in

$$-I_D R_S + V_{DS} - I_D R_D + V_{DD} = 0$$

and

$$\begin{aligned} V_{DS} &= -V_{DD} + I_D(R_D + R_S) \\ &= -20\text{ V} + (3.4\text{ mA})(2.7\text{ k}\Omega + 1.8\text{ k}\Omega) \\ &= -20\text{ V} + 15.3\text{ V} \\ &= \mathbf{-4.7\text{ V}} \end{aligned}$$

## 6.12 UNIVERSAL JFET BIAS CURVE

Since the dc solution of a FET configuration requires drawing the transfer curve for each analysis, a universal curve was developed that can be used for any level of $I_{DSS}$ and $V_P$. The universal curve for an *n*-channel JFET or depletion-type MOSFET (for negative values of $V_{GSQ}$) is provided in Fig. 6.58. Note that the horizontal axis is not that of $V_{GS}$ but of a normalized level defined by $V_{GS}/|V_P|$, the $|V_P|$ indicating that only the magnitude of $V_P$ is to be employed, not its sign. For the vertical axis, the scale is also a normalized level of $I_D/I_{DSS}$. The result is that when $I_D = I_{DSS}$, the ratio is 1, and when $V_{GS} = V_P$, the ratio $V_{GS}/|V_P|$ is $-1$. Note also that the scale for $I_D/I_{DSS}$ is on the left rather than on the right as encountered for $I_D$ in past exercises. The additional two scales on the right need an introduction. The vertical scale labeled *m* can in itself be used to find the solution to fixed-bias configurations. The other scale, labeled *M*, is employed along with the *m* scale to find the solution

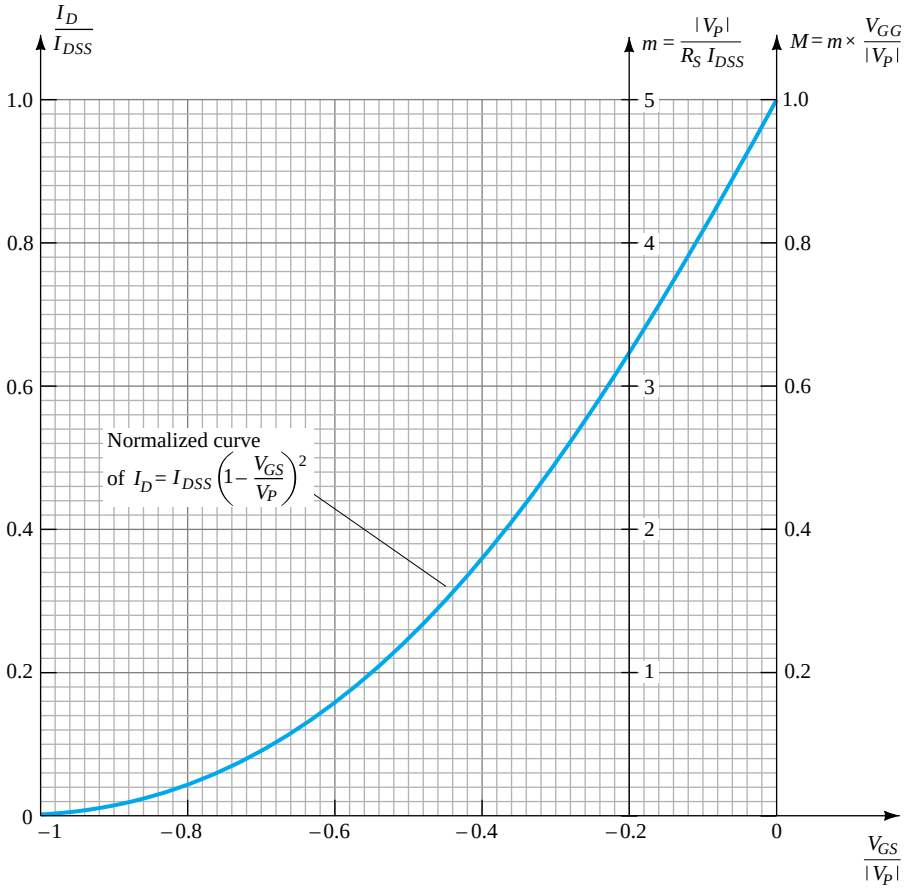



**Figure 6.58** Universal JFET bias curve.

to voltage-divider configurations. The scaling for $m$ and $M$ come from a mathematical development involving the network equations and normalized scaling just introduced. The description to follow will not concentrate on why the $m$ scale extends from 0 to 5 at $V_{GS}/|V_P| = -0.2$ and the $M$ scale from 0 to 1 at $V_{GS}/|V_P| = 0$ but rather on how to use the resulting scales to obtain a solution for the configurations. The equations for $m$ and $M$ are the following, with $V_G$ as defind by Eq. (6.15).

$$m = \frac{|V_P|}{I_{DSS}R_S} \tag{6.35}$$

$$M = m \times \frac{V_G}{|V_P|} \tag{6.36}$$

with $$V_G = \frac{R_2 V_{DD}}{R_1 + R_2}$$

Keep in mind that the beauty of this approach is the elimination of the need to sketch the transfer curve for each analysis, that the superposition of the bias line is a great deal easier, and that the calculations are fewer. The use of the $m$ and $M$ axes is best described by examples employing the scales. Once the procedure is clearly understood, the analysis can be quite rapid, with a good measure of accuracy.

*EXAMPLE 6.19* Determine the quiescent values of $I_D$ and $V_{GS}$ for the network of Fig. 6.59.

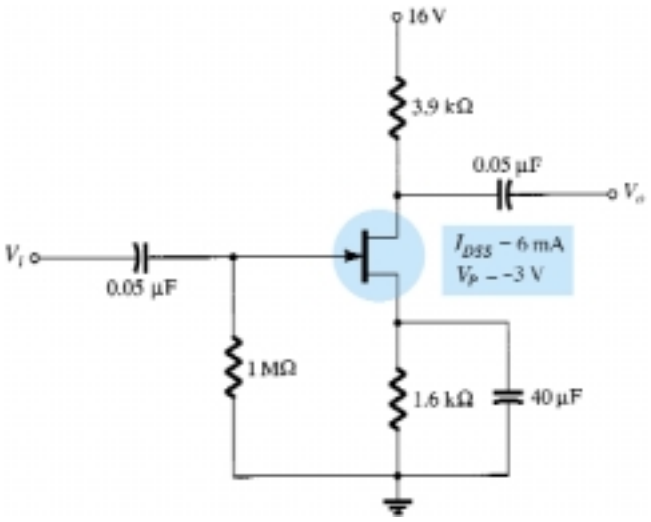



**Figure 6.59** Example 6.19.

## Solution

Calculating the value of $m$, we obtain

$$m = \frac{|V_P|}{I_{DSS}R_S} = \frac{|-3\text{ V}|}{(6\text{ mA})(1.6\text{ k}\Omega)} = 0.31$$

The self-bias line defined by $R_S$ is plotted by drawing a straight line from the origin through a point defined by $m = 0.31$, as shown in Fig. 6.60.
The resulting $Q$-point:

$$\frac{I_D}{I_{DSS}} = 0.18 \quad \text{and} \quad \frac{V_{GS}}{|V_P|} = -0.575$$

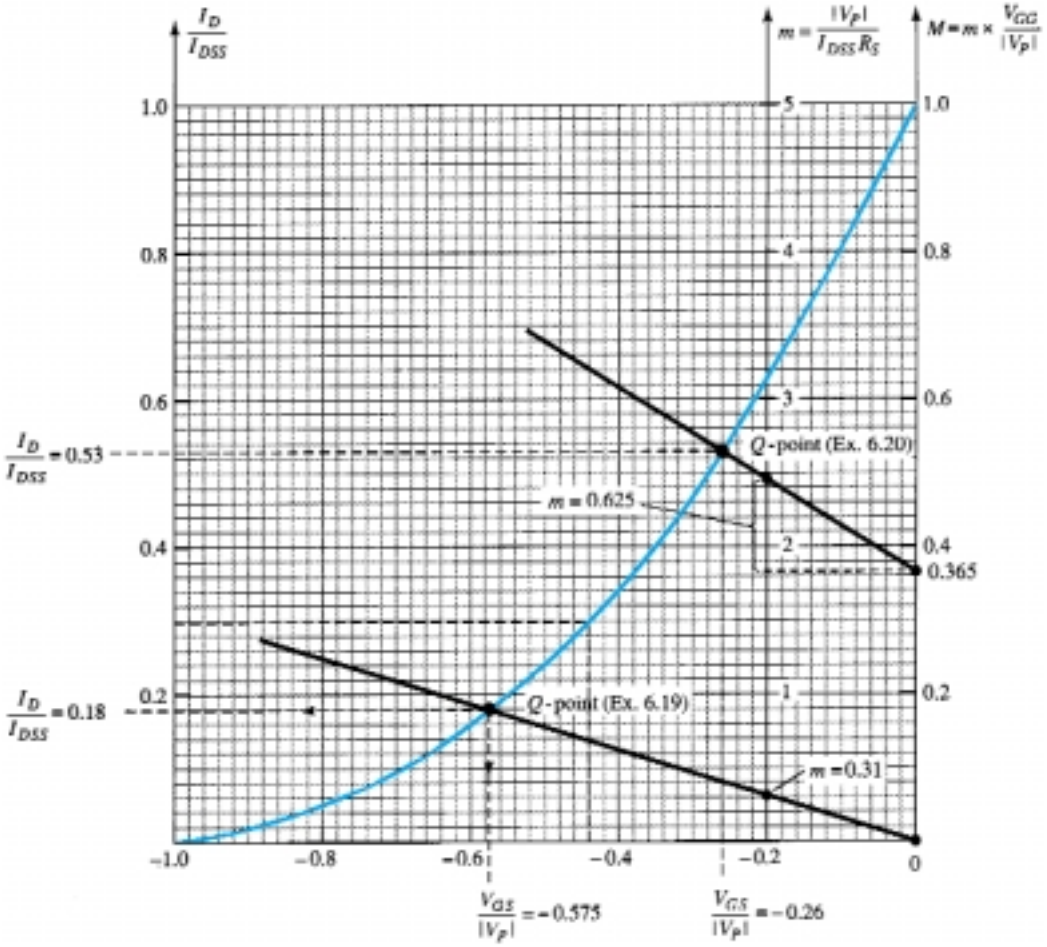



**Figure 6.60** Universal curve for Examples 6.19 and 6.20.

The quiescent values of $I_D$ and $V_{GS}$ can then be determined as follows:

$$I_{D_Q} = 0.18I_{DSS} = 0.18(6\text{ mA}) = \mathbf{1.08\text{ mA}}$$

and

$$V_{GS_Q} = -0.575|V_P| = -0.575(3\text{ V}) = \mathbf{-1.73\text{ V}}$$

---

*EXAMPLE 6.20*

Determine the quiescent values of $I_D$ and $V_{GS}$ for the network of Fig. 6.61.

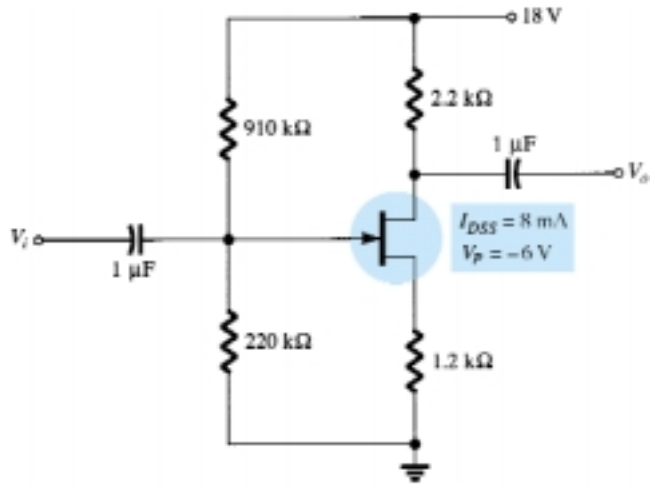



**Figure 6.61** Example 6.20.

### Solution

Calculating $m$ gives

$$m = \frac{|V_P|}{I_{DSS}R_S} = \frac{|-6\text{ V}|}{(8\text{ mA})(1.2\text{ k}\Omega)} = 0.625$$

Determining $V_G$ yields

$$V_G = \frac{R_2V_{DD}}{R_1 + R_2} = \frac{(220\text{ k}\Omega)(18\text{ V})}{910\text{ k}\Omega + 220\text{ k}\Omega} = 3.5\text{ V}$$

Finding $M$, we have

$$M = m \times \frac{V_G}{|V_P|} = 0.625\left(\frac{3.5\text{ V}}{6\text{ V}}\right) = 0.365$$

Now that $m$ and $M$ are known, the bias line can be drawn on Fig. 6.60. In particular, note that even though the levels of $I_{DSS}$ and $V_P$ are different for the two networks, the same universal curve can be employed. First find $M$ on the $M$ axis as shown in Fig. 6.60. Then draw a horizontal line over to the $m$ axis and, at the point of intersection, add the magnitude of $m$ as shown in the figure. Using the resulting point on the $m$ axis and the $M$ intersection, draw the straight line to intersect with the transfer curve and define the $Q$-point:

That is, $$\frac{I_D}{I_{DSS}} = 0.53 \quad \text{and} \quad \frac{V_{GS}}{|V_P|} = -0.26$$

and $$I_{DQ} = 0.53I_{DSS} = 0.53(8\text{ mA}) = \mathbf{4.24\text{ mA}}$$

with $$V_{GSQ} = -0.26|V_P| = -0.26(6\text{ V}) = \mathbf{-1.56\text{ V}}$$

## 6.13 PSPICE WINDOWS

### JFET Voltage-Divider Configuration

The results of Example 6.20 will now be verified using PSpice Windows. The network of Fig. 6.62 is constructed using computer methods described in the previous chapters. The J2N3819 JFET is obtained from the **EVAL.slb** library and, through **Edit-Model-Edit Instance Model (Text), Vto** is set to −6V and **Beta,** as defined by Beta = $I_{DSS}/|V_P|^2$ is set to 0.222 mA/V$^2$. After an **OK** followed by clicking the **Simulation** icon (the yellow background with the two waveforms) and clearing the **Message Viewer, PSpiceAD** screens will result in Fig. 6.62. The resulting drain cur-

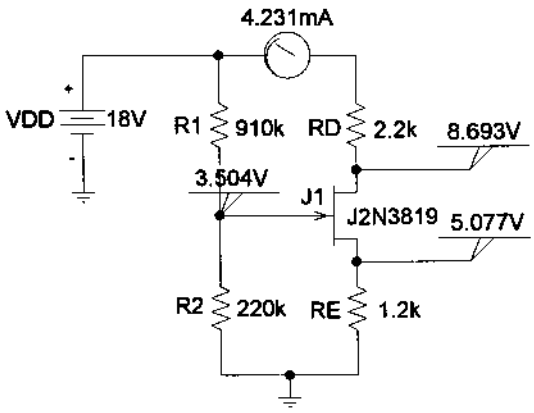



**Figure 6.62** JFET voltage-divider configuration with PSpice Windows results for the dc levels.

rent is 4.231 mA compared to the calculated level of 4.24 mA, and $V_{GS}$ is 3.504 V − 5.077 V = −1.573 V versus the calculated value of −1.56 V—both excellent comparisons.

## Combination Network

Next, the results of Example 6.13 with both a transistor and JFET will be verified. For the transistor, the **Model** must be altered to have a **Bf**(beta) of 180 to match the example, and for the JFET, **Vto** must be set to −6V and **Beta** to 0.333 mA/V$^2$. The results appearing in Fig. 6.63 are again an excellent comparison with the hand-written solution. $V_D$ is 11.44 V compared to 11.07 V, $V_C$ is 7.138 V compared to 7.32 V, and $V_{GS}$ is −3.758 V compared to −3.7 V.

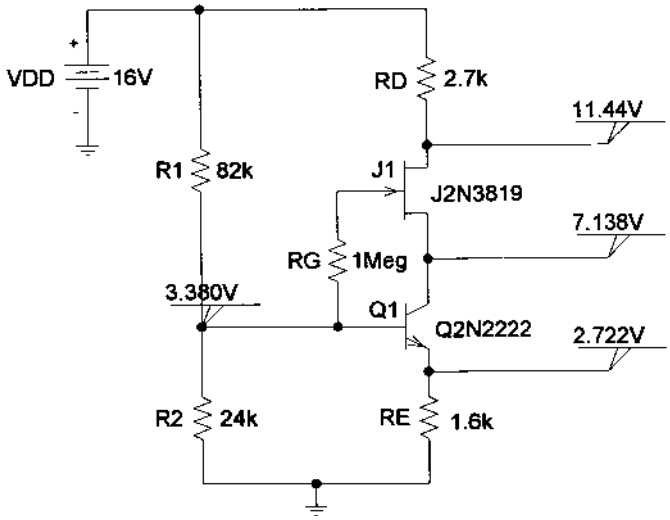



**Figure 6.63** Verifying the hand-calculated solution of Example 6.13 using PSpice Windows.

## Enhancement MOSFET

Next, the analysis procedure of Section 6.6 will be verified using the IRF150 enhancement-type *n*-channel MOSFET found in the **EVAL.slb** library. First, the device characteristics will be obtained by constructing the network of Fig. 6.64.

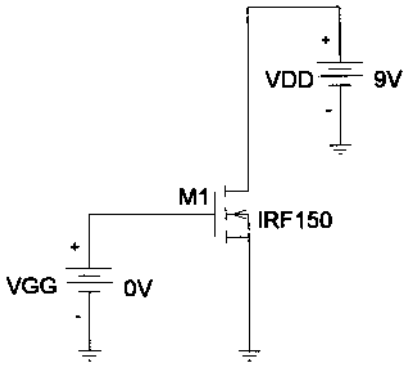



**Figure 6.64** Network employed to obtain the characteristics of the IRF150 enhancement-type *n*-channel MOSFET.

Clicking on the **Setup Analysis** icon (with the blue bar at the top in the left-hand corner of the screen), **DC Sweep** is chosen to obtain the **DC Sweep** dialog box. **Voltage Source** is chosen as the **Swept Var. Type,** and **Linear** is chosen for the **Sweep Type.** Since only one curve will be obtained, there is no need for a **Nested Sweep.** The voltage-drain voltage VDD will remain fixed at a value of 9 V (about three times the threshold value (**Vto**) of 2.831 V), while the gate-to-source voltage $V_{GS}$, which in

this case is VGG, will be swept from 0 to 10 V. The **Name** therefore is VGG and the **Start Value** 0V, the **End Value** 10V, and the **Increment** 0.01V. After an **OK** followed by a **Close** of the **Analysis Setup,** the analysis can be performed through the **Analysis** icon. If **Automatically run Probe after simulation** is chosen under the **Probe Setup Options** of **Analysis,** the **OrCAD-MicroSim Probe** screen will result, with the horizontal axis appearing with VGG as the variable and range from 0 to 10 V. Next, the **Add Traces** dialog box can be obtained by clicking the **Traces** icon (red pointed pattern on an axis) and the **ID(M1)** chosen to obtain the drain current versus the gate-to-source voltage. Click **OK,** and the characteristics will appear on the screen. To expand the scale of the resulting plot to 20 V, simply choose **Plot** followed by **X-Axis Settings** and set the **User Defined** range to 0 to 20 V. After another **OK,** and the plot of Fig. 6.65 will result, revealing a rather high-current device. The labels **ID** and **VGS** were added using the **Text Label** icon with the letters A, B, and C. The hand-drawn load line will be described in the paragraph to follow.

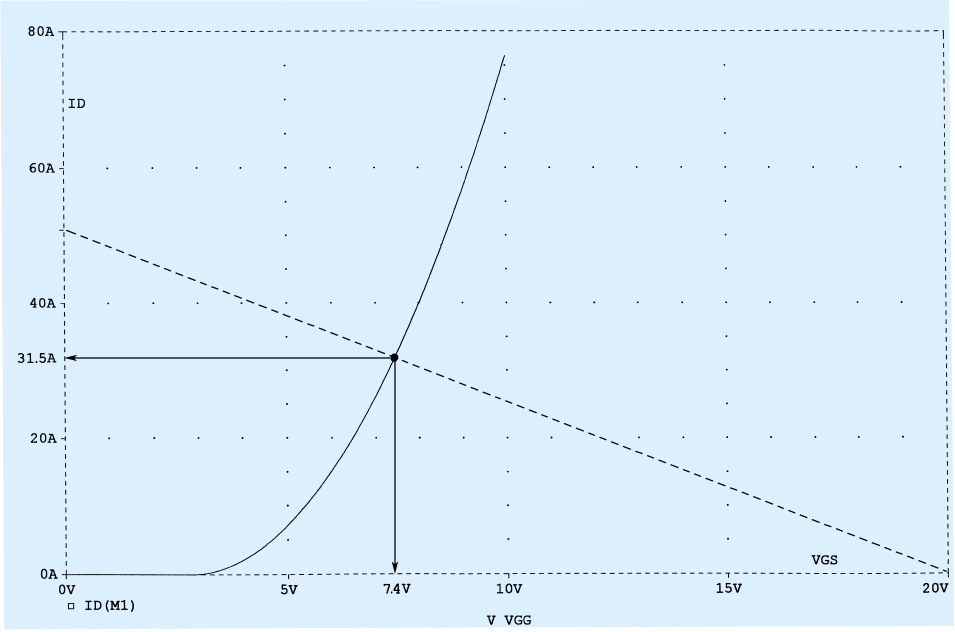



**Figure 6.65** Characteristics of the IRF500 MOSFET of Figure 6.64 with a load line defined by the network of Figure 6.66.

The network of Fig. 6.66 was then established to provide a load line extending from $I_D$ equal to 20 V/0.4 Ω = 50 A down to $V_{GS} = V_{GG} = 20$ V as shown in Fig. 6.65. A simulation resulted in the levels shown, which match the solution of Fig. 6.65.

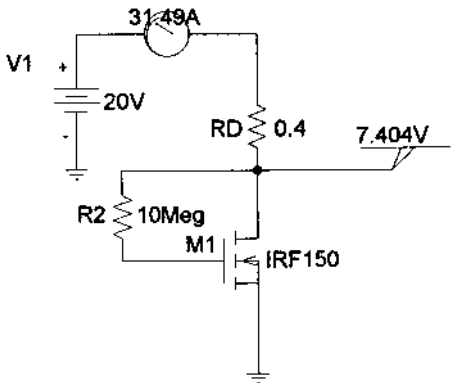



**Figure 6.66** Feedback-biasing arrangement employing an IRF150 enhancement-type MOSFET.

## § 6.2 Fixed-Bias Configuration

**1.** For the fixed-bias configuration of Fig. 6.67:
(a) Sketch the transfer characteristics of the device.
(b) Superimpose the network equation on the same graph.
(c) Determine $I_{D_Q}$ and $V_{DS_Q}$.
(d) Using Shockley's equation, solve for $I_{D_Q}$ and then find $V_{DS_Q}$. Compare with the solutions of part (c).

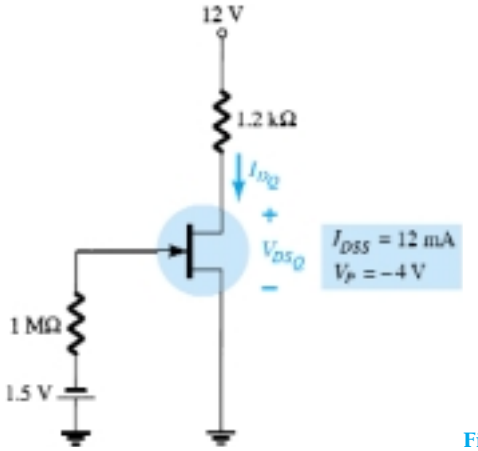



**Figure 6.67** Problems 1, 35

**2.** For the fixed-bias configuration of Fig. 6.68, determine:
(a) $I_{D_Q}$ and $V_{GS_Q}$ using a purely mathematical approach.
(b) Repeat part (a) using a graphical approach and compare results.
(c) Find $V_{DS}$, $V_D$, $V_G$, and $V_S$ using the results of part (a).

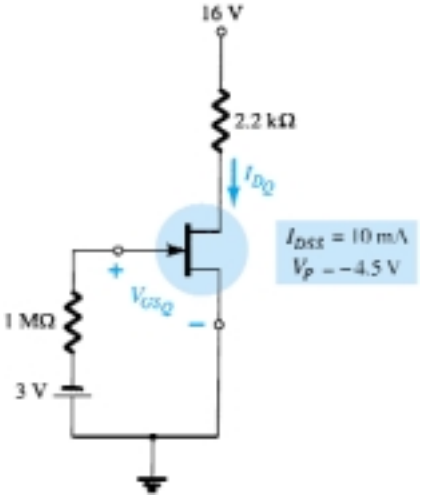



**Figure 6.68** Problem 2

**3.** Given the measured value of $V_D$ in Fig. 6.69, determine:
(a) $I_D$.
(b) $V_{DS}$.
(c) $V_{GG}$.

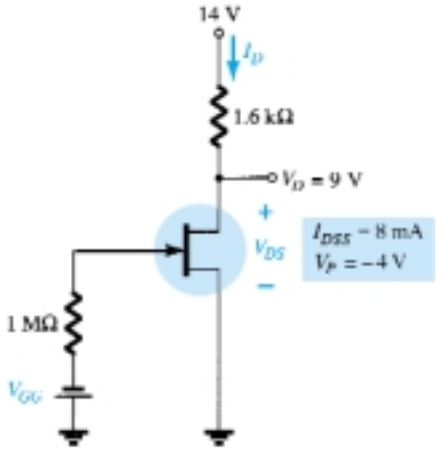



**Figure 6.69** Problem 3

4. Determine $V_D$ for the fixed-bias configuration of Fig. 6.70.

5. Determine $V_D$ for the fixed-bias configuration of Fig. 6.71.

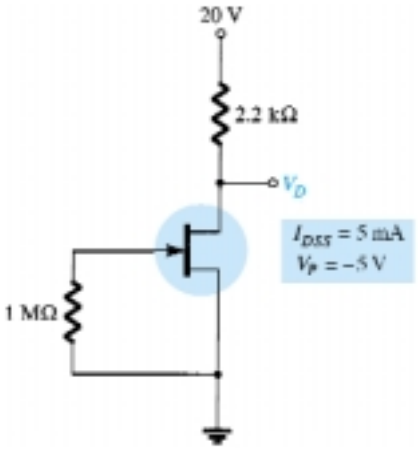



**Figure 6.70** Problem 4

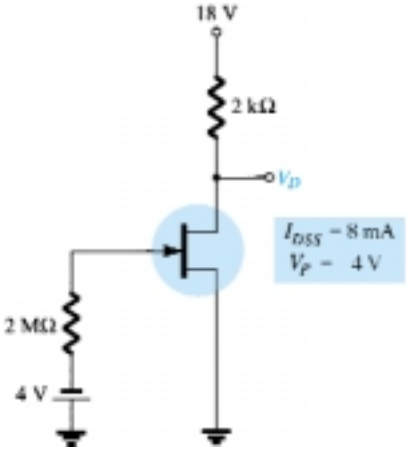



**Figure 6.71** Problem 5

## § 6.3 Self-Bias Configuration

6. For the self-bias configuration of Fig. 6.72:
   (a) Sketch the transfer curve for the device.
   (b) Superimpose the network equation on the same graph.
   (c) Determine $I_{DQ}$ and $V_{GS_Q}$.
   (d) Calculate $V_{DS}$, $V_D$, $V_G$, and $V_S$.

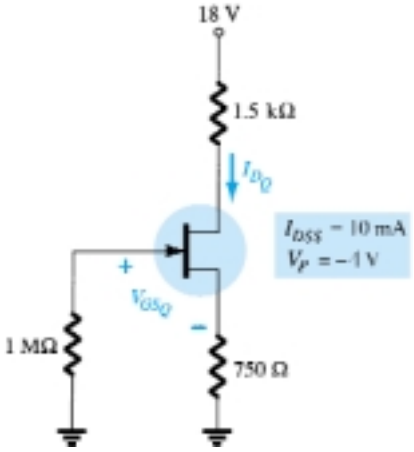



**Figure 6.72** Problems 6, 7, 36

* 7. Determine $I_{D_Q}$ for the network of Fig. 6.72 using a purely mathematical approach. That is, establish a quadratic equation for $I_D$ and choose the solution compatible with the network characteristics. Compare to the solution obtained in Problem 6.

8. For the network of Fig. 6.73, determine:
   (a) $V_{GS_Q}$ and $I_{D_Q}$.
   (b) $V_{DS}$, $V_D$, $V_G$, and $V_S$.

9. Given the measurement $V_S = 1.7$ V for the network of Fig. 6.74, determine:
   (a) $I_{D_Q}$.
   (b) $V_{GS_Q}$.
   (c) $I_{DSS}$.
   (d) $V_D$.
   (e) $V_{DS}$.

* **10.** For the network of Fig. 6.75, determine:
(a) $I_D$.
(b) $V_{DS}$.
(c) $V_D$.
(d) $V_S$.

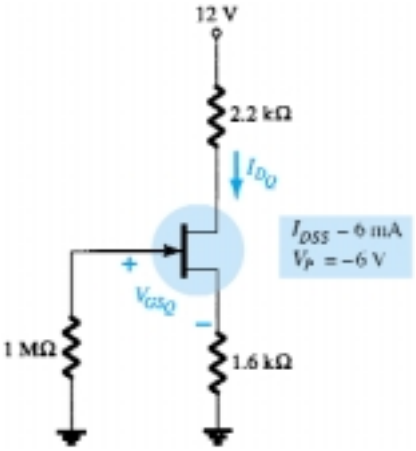



**Figure 6.73** Problem 8

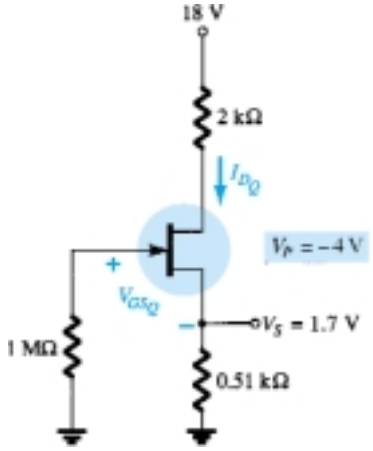



**Figure 6.74** Problem 9

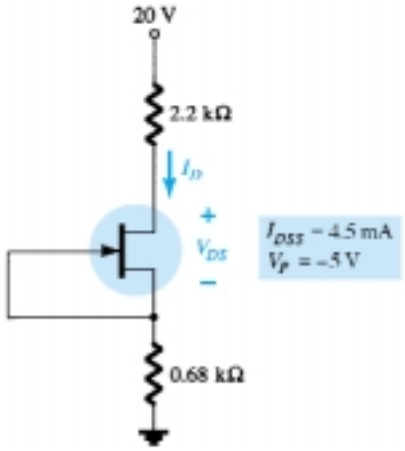



**Figure 6.75** Problem 10

* **11.** Find $V_S$ for the network of Fig. 6.76.

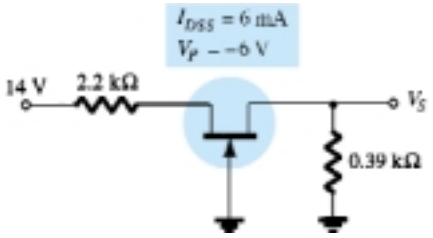



**Figure 6.76** Problem 11

## § 6.4 Voltage-Divider Biasing

**12.** For the network of Fig. 6.77, determine:
(a) $V_G$.
(b) $I_{D_Q}$ and $V_{GS_Q}$.
(c) $V_D$ and $V_S$.
(d) $V_{DS_Q}$.

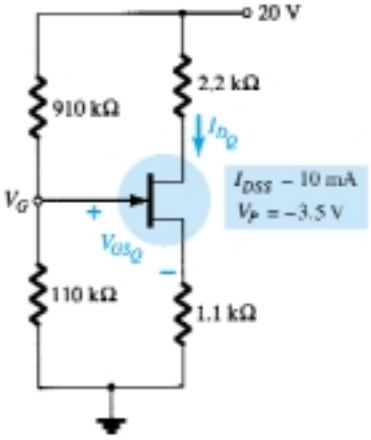



**Figure 6.77** Problems 12, 13

**13.** (a) Repeat Problem 12 with $R_S = 0.51$ kΩ (about 50% of the value of 12). What is the effect of a smaller $R_S$ on $I_{D_Q}$ and $V_{GS_Q}$?
(b) What is the minimum possible value of $R_S$ for the network of Fig. 6.77?

**14.** For the network of Fig. 6.78, $V_D = 9$ V. Determine:
(a) $I_D$.
(b) $V_S$ and $V_{DS}$.
(c) $V_G$ and $V_{GS}$.
(d) $V_P$.

* **15.** For the network of Fig. 6.79, determine:
(a) $I_{D_Q}$ and $V_{GS_Q}$.
(b) $V_{DS}$ and $V_S$.

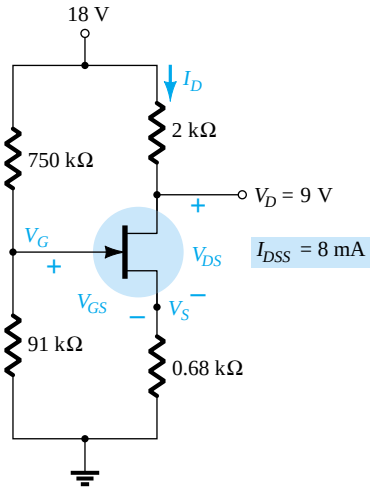



**Figure 6.78** Problem 14

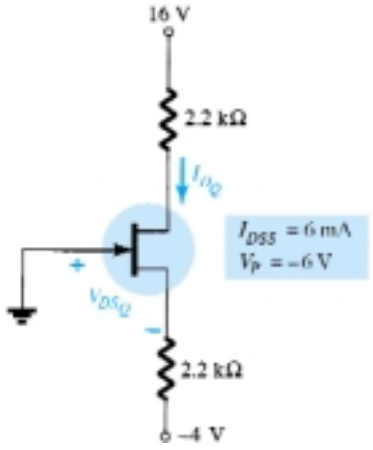



**Figure 6.79** Problems 15, 37

* **16.** Given $V_{DS} = 4$ V for the network of Fig. 6.80, determine:
(a) $I_D$.
(b) $V_D$ and $V_S$.
(c) $V_{GS}$.

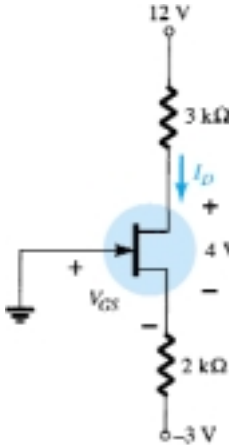



**Figure 6.80** Problem 16

## § 6.5 Depletion-Type MOSFETs

**17.** For the self-bias configuration of Fig. 6.81, determine:
(a) $I_{D_Q}$ and $V_{GS_Q}$.
(b) $V_{DS}$ and $V_D$.

* **18.** For the network of Fig. 6.82, determine:
(a) $I_{D_Q}$ and $V_{GS_Q}$.
(b) $V_{DS}$ and $V_S$.

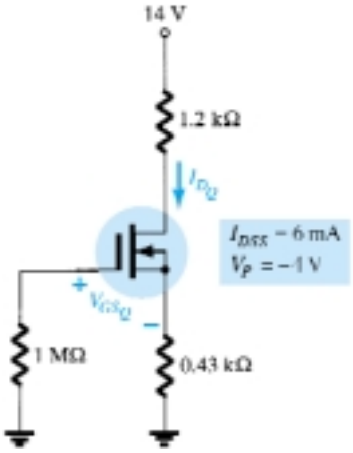



**Figure 6.81** Problem 17

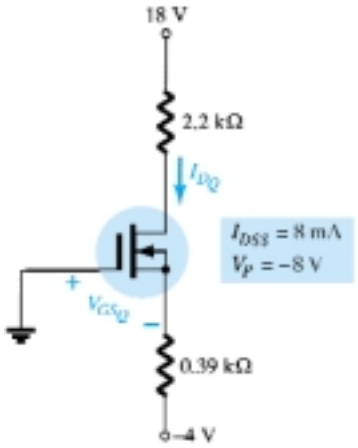



**Figure 6.82** Problem 18

## § 6.6 Enhancement-Type MOSFETs

**19.** For the network of Fig. 6.83, determine:
(a) $I_{D_Q}$.
(b) $V_{GS_Q}$ and $V_{DS_Q}$.
(c) $V_D$ and $V_S$.
(d) $V_{DS}$.

**20.** For the voltage-divider configuration of Fig. 6.84, determine:
(a) $I_{D_Q}$ and $V_{GS_Q}$.
(b) $V_D$ and $V_S$.

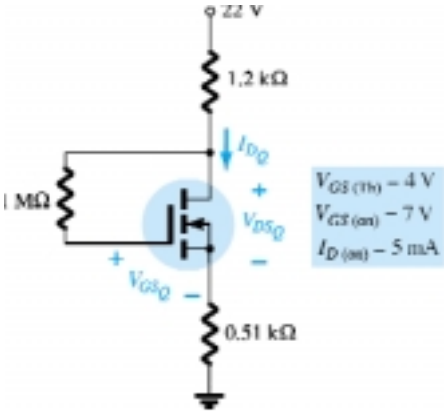



**Figure 6.83** Problem 19

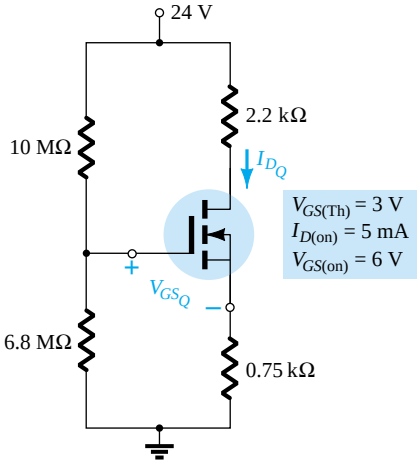



**Figure 6.84** Problem 20

## § 6.8 Combination Networks

* **21.** For the network of Fig. 6.85, determine:
(a) $V_G$.
(b) $V_{GS_Q}$ and $I_{D_Q}$.
(c) $I_E$.
(d) $I_B$.
(e) $V_D$.
(f) $V_C$.

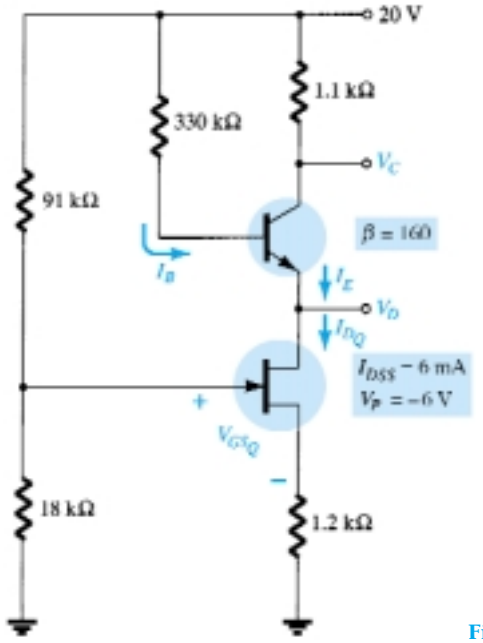



**Figure 6.85** Problem 21

* **22.** For the combination network of Fig. 6.86, determine:
  (a) $V_B$ and $V_G$.
  (b) $V_E$.
  (c) $I_E$, $I_C$, and $I_D$.
  (d) $I_B$.
  (e) $V_C$, $V_S$, and $V_D$.
  (f) $V_{CE}$.
  (g) $V_{DS}$.

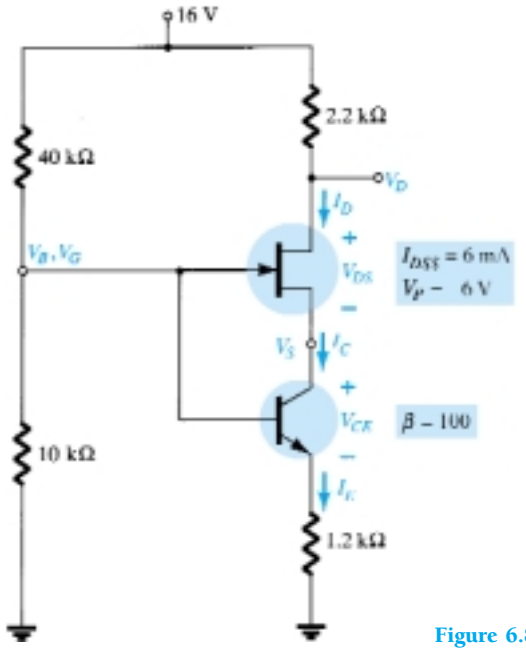



**Figure 6.86** Problem 22

## § 6.9 Design

* **23.** Design a self-bias network using a JFET transistor with $I_{DSS} = 8$ mA and $V_P = -6$ V to have a $Q$-point at $I_{D_Q} = 4$ mA using a supply of 14 V. Assume that $R_D = 3R_S$ and use standard values.

* **24.** Design a voltage-divider bias network using a depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V to have a $Q$-point at $I_{D_Q} = 2.5$ mA using a supply of 24 V. In addition, set $V_G = 4$ V and use $R_D = 2.5R_S$ with $R_1 = 22$ MΩ. Use standard values.

**25.** Design a network such as appears in Fig. 6.39 using an enhancement-type MOSFET with $V_{GS(Th)} = 4$ V, $k = 0.5 \times 10^{-3}$A/V$^2$ to have a $Q$-point of $I_{D_Q} = 6$ mA. Use a supply of 16 V and standard values.

## § 6.10 Troubleshooting

* **26.** What do the readings for each configuration of Fig. 6.87 suggest about the operation of the network?

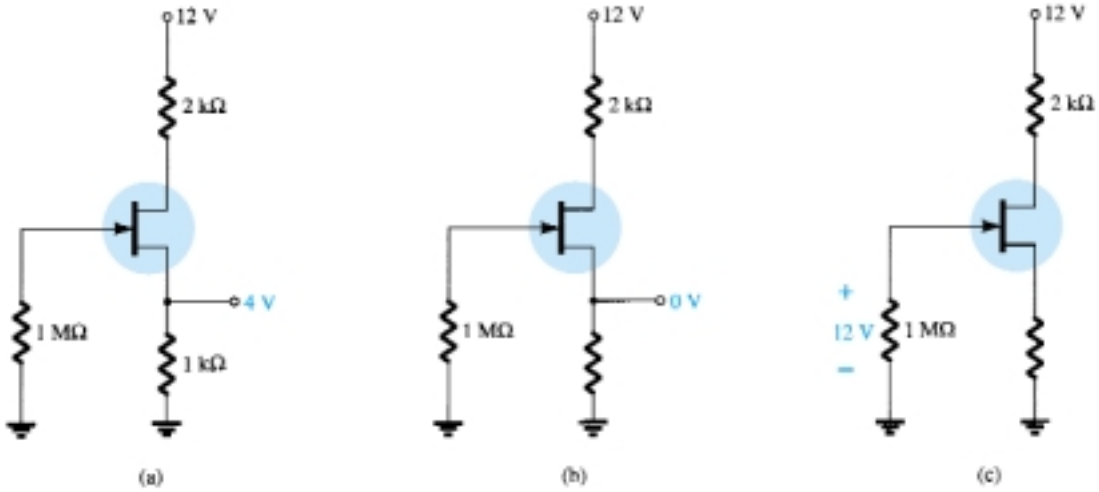



**Figure 6.87** Problem 26

* **27.** Although the readings of Fig. 6.88 initially suggest that the network is behaving properly, determine a possible cause for the undesirable state of the network.

* **28.** The network of Fig. 6.89 is not operating properly. What is the specific cause for its failure?

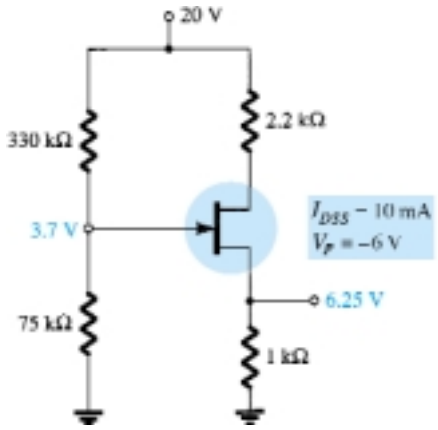



**Figure 6.88** Problem 27

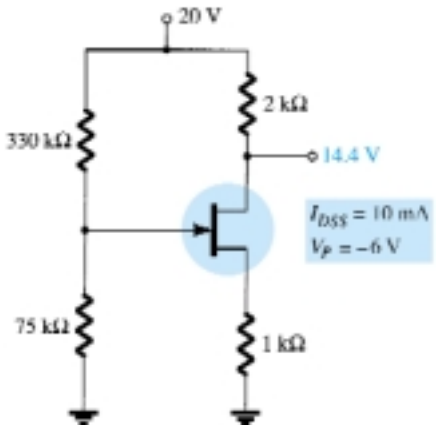



**Figure 6.89** Problem 28

## § 6.11 *p*-Channel FETs

**29.** For the network of Fig. 6.90, determine:
(a) $I_{D_Q}$ and $V_{GS_Q}$.
(b) $V_{DS}$.
(c) $V_D$.

**30.** For the network of Fig. 6.91, determine:
(a) $I_{D_Q}$ and $V_{GS_Q}$.
(b) $V_{DS}$.
(c) $V_D$.

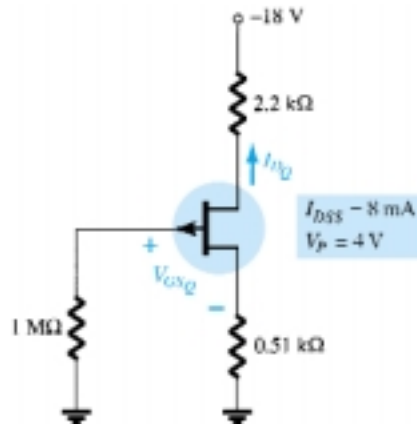



**Figure 6.90** Problem 29

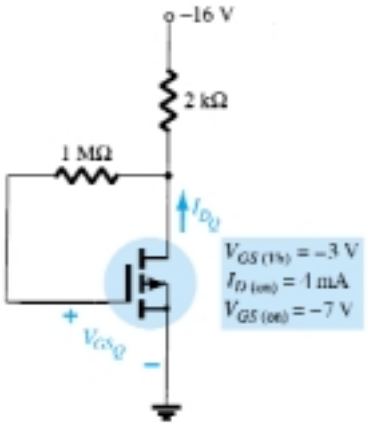



**Figure 6.91** Problem 30

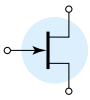
## § 6.12 Universal JFET Bias Curve

**31.** Repeat Problem 1 using the universal JFET bias curve.

**32.** Repeat Problem 6 using the universal JFET bias curve.

**33.** Repeat Problem 12 using the universal JFET bias curve.

**34.** Repeat Problem 15 using the universal JFET bias curve.



## § 6.13 PSpice Windows

**35.** Perform a PSpice Windows analysis of the network of Problem 1.

**36.** Perform a PSpice Windows analysis of the network of Problem 6.

**37.** Perform a PSpice Windows analysis of the network of Problem 15.

*Please Note: Asterisks indicate more difficult problems.